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NSP/004/041 - Specification for the construction of LV ABC Overhead Lines

1. Purpose

The purpose of this document is to provide a specification detailing the design and construction requirements for new low voltage ABC (Aerial Bundled Conductor) distribution systems. Additionally this specification confirms the reference OHL6/10 as the reference to be quoted on all planning applications for new lines or on correspondence with the Department of Energy and Climate Change.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/041	3 (Oct 09)	Specification for LV ABC Overhead Lines
NSP/004/041	2 (Jan 07)	Specification for LV ABC Overhead Lines
NSP/004/041	1 (May 06)	Specification for LV ABC Overhead Lines
DSS/004/041	1.1	Specification for LV ABC Overhead Lines in Yorkshire Electricity
O.305	Feb 1998	Low Voltage Overhead Distribution Systems

2. Scope

This specification covers the application policy and constructional requirements for new LV ABC three phase distribution lines supported on poles or attached to buildings. It has been designed in accordance with ENA TS 43-12, 43-13 and 43-14 utilising 4 x 70mm², 4 x 120mm² both with and without protective conductors or a 5th core for main lines and 4 x 35mm² and 2 x 35mm² for service lines.

Renovation of existing open wire networks to ABC construction and additions of conductors to existing open wire networks are detailed in the supporting document NSP/004/041/001.

New overhead line services and new or replacement surface wiring will be carried out in accordance with this document and NSP/004/043.

2.1 Application Policy

ABC shall be used for the following:

- Construction of all new LV overhead lines
- The extension or the selected refurbishment of existing LV overhead lines
- As a means of providing supplies formally achieved via surface wiring or eaves mains see NSP/004/043.
- In areas identified as high risk by DSS/023 "Risk Assessment of Overhead Lines"
- Area's identified in NSP/004/011 "Guidance on Overhead Line Clearances" as being not suitable for "non effectively insulated conductors"

Main Lines: 4 x 120mm² with and without protective conductors or 4 x 70mm²

Service Lines: 4 x 35mm² or 2 x 35mm²

Additional information has been included to cover work or extensions to existing 4 x 50 & 4 x 95mm² sizes.

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3 Technical Specification

3.1 Introduction

ABC is a fully insulated, 4 wire LV distribution system. Four identical compacted stranded aluminium conductors are insulated with black XLPE and twisted into a bundled formation. The bundle is erected and secured as though it were a single overhead line conductor, supported and terminated in suspension or anchor clamps which grip the outer insulation of all four conductors.

All ABC overhead lines shall be constructed in accordance with the provisions of this document. This system is designed for use in association with Protective Multiple Earthing, however where the application of the company's earthing policy IMP/010/011 requires the installation of a separate protective conductor this shall be provided through the use of a 5 core ABC that includes within the bundle a 25mm² protective conductor.

3.2 Design Criteria

3.2.1 Weather Loadings

Lines to this specification have been designed with the following weather loadings:

Wind Pressure 380N/m²
Diametric Ice 9.5mm

3.2.2 Conductors

This specification applies the loading conditions detailed in clause 3.2.1 and includes the following assumptions:

1. The ultimate tensile strength of the aluminium conductor is 170 N.mm².
2. The average elastic limit of extension for the conductor material is 60 per cent of the ultimate tensile strength.

This specification uses a maximum design stress of 50 per cent of the elastic stress limit for service lines and 40 percent in the case of four core main lines. In both service and main lines, compliance with these stress levels occurs with the smallest ABC. (This effectively provides a factor of safety of at least four on the UTS of the conductor. These conditions permit reasonable access to the cores for the application of connections.

Appendix 1 & 1A provides details of the technical data used in preparing sag/tension and pole/stay selection information.

The tension of each size of ABC under design loading conditions is used for all stay and pole selection data. These figures having been derived using actual MWT values for each conductor size in place of the common mass/tension ratios at 10 degrees Celsius used in ENA TS 43-12.

Note 1

4 x 120mm² ABC did not form part of the original ENA TS 43-13 specification. However loading and impedance requirements necessitated the introduction of a larger size. Although 4 x 120mm² ABC is considerably stronger than 4 x 95mm² ABC, MWT and erection tensions have been designed to match those of 4 x 95mm² ABC. (Any route suitable for 4 x 95mm² ABC will also be suitable for 4 x 120mm² ABC).

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Note 2

The effect of adding a 25 mm² protective conductor to the 4 x 120mm² ABC has negligible effect on the mechanical loadings, hence no additional sag charts or stay charts have been added to this specification.

3.2.3 Supports

The stresses created in unstayed intermediate supports are considered as bending stresses due to wind load on iced conductors and the theoretical equivalent force created by two service span attachments erected in line with the normal route of the distribution line. The loading point for the forces has been taken as 300mm below the pole top. This height has been chosen to provide a balanced transition between new and existing poles that have been previously fabricated for open wire conductors.

The unstayed capability of supports used within this specification has been designed to provide a minimum factor of safety of 2.5 for all conductor sizes.

The stresses created in stayed supports are crippling stresses due to stay tension and conductor weight.

The strut capabilities of supports used within this specification have been designed to provide a minimum factor of safety of 2.5 for all conductor sizes.

The largest pole which has been considered within the support loading calculations is 17m, with a wind loading span of 90m. If longer poles are required then these situations shall be referred to the Overhead Line Standards Manager.

3.2.4 Sag Tables

ABC Mains and Services shall be erected to the Sag Tables shown in Appendix 3, 4, 5, 6 and 7.

Appendix 3 - Sag Table for 35mm² ABC

Appendix 4 - Sag Table for 50mm² ABC (info purposes only)

Appendix 5 - Sag Table for 70mm² ABC

Appendix 6 - Sag Table for 95mm² ABC (info purposes only)

Appendix 7 - Sag Table for 120mm² ABC (with or without 25 mm² protective conductor)

Appendix 8 - Sag Table for Slack Spans

3.2.5 Span Lengths

Mains (70 & 120mm² ABC)

Basic Design Span	50m *
Maximum Span	90m
Maximum Wind Loading Span	90m
Average Span	50m

Services (2 and 4 core 35mm² ABC)

Basic Span	50m	between wood
Maximum Span	70m	pole supports
Basic Span	20m	when attached
Maximum Span	30m	to a building

* Notes

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The 50m basic span has been selected on the assumption that pole positions are normally dictated by the need for service spans to properties supplied from the main LV Main. Where no services spans are required, especially on rural LV network, users are reminded that the spanning can be increased closer to the stated 90m spanning to maximise the design capability of the route.

3.2.6 Erection Data

Main lines shall be erected in accordance with Appendices 3 - 7.

Spans attached to buildings shall be erected at reduced tensions in accordance with Appendix 8.

Erection sag tables are based upon the requirements that:-

- (a) Conductor clamps shall not exceed a load beyond which damage may occur to the conductor insulation, and in any event shall not exceed 40% of the breaking load of any clamp.
- (b) Conductor tension shall be such that under all reasonable working conditions, the bundle may be separated for making connections.
- (c) Conductor tension is high enough to ensure efficient operation of anchor clamps throughout the entire range of design temperatures, whilst also ensuring the minimum sag for reasons of clearance.
- (d) 2 x 35mm² single phase ABC service lines are designed so that they can be uprated to 4 x 35mm² three phase service lines without further structural alterations.

Maximum working tensions are:-

- 11.4 kN for 4 x 95 or 4 x 120mm² ABC – Mains (with or without 25 mm² protective conductor)
- 9.04 kN for 4 x 70mm² ABC - Mains
- 7.29 kN for 4 x 50mm² ABC - Mains (historical info purposes only)
- 4.2 kN for 4 x 35mm² ABC - Long services lines not connected directly to a building
- 2.7 kN for 2 x 35mm² ABC - Long service lines not connected directly to a building
- 1.3 kN for all sizes of ABC - Where a service is attached directly to a building

3.2.7 Clearances

The minimum height of all line conductors and clearances to other objects shall be compliant with the current version of NSP/004/011. The above document supplements the basic clearance requirements as recommended by ENA TS 43-8 "Overhead Line Clearances" with additional company requirements.

For the purposes of evaluating clearances, the sags at the maximum operating temperature of 75°C shall be used.

3.3 Technical requirements

3.3.1 Systems attached to Poles

Systems attached to poles should be generally in accordance with the arrangements illustrated in Drawing Nos. 1000431202 to 1000431223.

All fittings shall provide an insulating barrier between the core insulation and the mechanical attachment. The ABC shall be attached in such a manner that it does not make direct or inadvertent contact with any metalwork or stays. When attached to steel poles, the ABC shall not be placed where it may make direct or inadvertent contact with the pole.

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This standard provides for line deviations of up to 90°, with intermediate supports being limited to 60°.

3.3.2 Systems attached to Buildings

Systems attached to buildings shall comply with the requirements of Drawing Nos. 1000431219 and 1000431220. Account must be taken of the mechanical loadings on the building due to the attachment, which should not exceed 1.3 kN (130kg) per fixing, unless special precautions are taken and no system shall be constructed with full aerial tensions acting directly on buildings. All fixings should preferably be loaded in shear and not in tension. Angles of approach to buildings should take advantage of corners or structural features such that the angle to the building surface under load is minimised. Appendix 9 provides sag tension data, which complies with a design loading for the building attachments of 1.3 kN.

Where 4 x 35mm² ABC is used as a three phase overhead service, it may be run between wood pole supports in a similar fashion to the ABC main and terminated at a building by means of a wall bracket. Where this occurs, the conductor tension in the last span shall be reduced as detailed in the specification.

ABC attached to the face of a building commonly described as eaves main or surface wiring, shall be secured with fixing devices as specified in Drawing No. 1000431409 to provide a minimum spacing of 10 mm between the core insulation and the surface of the building. The frequency of cleating at a maximum interval of 1000 mm shall be such that the ABC when erected shall not be allowed to touch the fabric of the building or be attached in such a manner that the ABC may inadvertently contact the building, after erection. Routing of ABC shall take into account potential points of hazard to the installed system.

Where ABC is required to pass in close proximity (less than 50 mm) to any metallic pipework or cables not part of the company's network, the ABC shall be secured against contact or alternatively be protected from contact by means of additional protection as detailed in clause 3.10.2, 'Associated Materials'.

Where ABC is adjacent to an opening in the building, such as a window, and there is a possibility of interference by persons from within the building, then additional proximity protection as detailed in clause 3.10.2 should be fitted. Typical areas where consideration has to be given to fitting proximity protection are shown on Drawing No. 1000431219.

Where the attachment of ABC above 2.4 metres from ground level is such that there is a potential risk of mechanical damage by the movement of vehicles, or by the placing of materials/equipment, the ABC shall be additionally protected for mechanical purposes by means of a suitable non-conducting cable guard, as detailed in Clause 3.10.2

Additional guidance associated with the provision of eaves main or surface wiring can be found in NSP/004/043.

3.4 Supports

3.4.1 Intermediate supports

Intermediate supports, with or without aerial services, shall be constructed using wood poles of medium grade for all sizes of ABC main in accordance with Chart 6a of ENA TS 43-12. This table shall also be used to represent the capabilities of 4 x 120mm² ABC main.

The supports shall be configured in accordance with the following arrangement drawings

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Arrangement	Drawing Number
In-Line Intermediate	1000431202
Angle support up to 30° Deviation	1000431203
Inside angle support up to 30° Deviation	1000431204
Angle support up to 60° Deviation	1000431205

In all cases the hook bolt shall be arranged to bisect the line angle. Line deviations of up to 60° may be accommodated with a single roller suspension clamp and extension rollers must be fitted during conductor erection on angles in excess of 30°.

With the exception of unstayed angle poles, all support foundations shall be installed without kicking blocks at the depths shown:

Pole Height	Pole Sinking Depth
9.0 - 10.0m	1.6m
11.0 - 14.0m	1.8m

3.4.2 Unstayed angles on intermediate supports

Unstayed angle supports shall be avoided wherever possible. However they are permitted providing they comply with the following criteria:

ABC Main – On Medium grade poles the max angle of deviation does not exceed 7°

ABC Main - On Stout grade poles the max angle of deviation does not exceed 19°

Extended Service Span - Medium grade pole with max angle of deviation 30°

In all cases the max pole height is limited to 12m and the poles must be sunk to a depth of 1.8m and fitted with 2 x ENA TS 43-91 Type 2 (1300 x 250 x 125)mm blocks located 500mm below ground level and attached with a M20 bolts. Where access and ground types are suitable, augured foundations may be used as an alternative to blocked foundations. See drawing 1090431216 for foundation details.

Tubular steel poles of minimum class 'K' may be utilised to provide unstayed angles in excess of the above limits. Their use should be limited due to the excessive cost and protracted installation requirements.

3.4.3 Service attachments on unstayed angles

The pole selection shown in clause 3.4.1 allows for the equivalent of two service span attachments loading the support additionally to the angle of deviation. Services in excess of two may be connected provided

- The angles of connection give resultant forces that cancel each other out
- The pole classification is stout

3.4.4 Stayed supports

Stayed supports shall typically be of medium grade and be designed to be in accordance with clause 3.5 and the minimum stay spread tables detailed in Appendix 9

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3.4.5 Section Supports

Section pole arrangements shall supported on minimum class medium grade poles for all ABC conductor sizes provided the pole is a stayed support. Details of minimum stay angles can be found in appendix 9.

Arrangement	Drawing Number
Section Support, Angles 20° to 90° Line Deviation	1000431206
Section support, including section fuses	1000431207
Section support, transition pole	1000431208

Transition supports between existing open wire networks and ABC networks shall be arranged as shown on drawing number 1000431208. Details on the requirements for out of balance stays between the two design systems can be found in appendix 10.

3.4.6 Terminal Supports

Terminal arrangements shall be supported on minimum medium grade poles for all ABC conductor sizes. Details of minimum stay angles can be found in appendix 9. Terminal supports shall be arranged in accordance with the following drawing numbers.

Arrangement	Drawing Number
Terminal support	1000431209
Terminal support with underground cable termination	1000431210
Tee off from intermediate support	1000431211
Tee off from section support	1000431212

Tee-off poles shall be a minimum grade stout.

3.4.7 Tubular Steel Supports

New tubular steel supports designed in accordance with drawing number 1091193501 sht 2 shall be erected on a concrete paving stone located 1675mm below ground level then encased with between 100-300mm of concrete depending upon the pole type. See drawing 1091193502 sht 1 for foundation details. Additional information has been provided in the renovation document NSP/004/041/001.

3.5 Stays

All stay strands and components shall be in accordance with ENA TS 43-91.

Stay strands shall be 7/4.00mm (grade 1150) in accordance with ENA TS 43-91 Table 1 providing a nominal minimum failing load of 101 kN and all stay rods shall be Type 2 (18mm dia) as detailed on drawing 1000439109.

All stays shall include a single Type 1 stay insulator in accordance with drawing no 1000439107.

Stay spreads will normally be such that an angle of 45° is provided between stay and the vertical face of the pole. As a general rule, therefore, the stay spread on horizontal ground will be equal to the length of the pole less its sinking depth. In circumstances when tighter stay spreads are unavoidable because of the location of physical obstructions, reduced values in line with the data provided in the appendix 9 shall be used. In all cases the minimum allowed stay angle should be 20°.

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Stay blocks shall be installed at a standard installation depth of 1.8m irrespective of the stay spread. Where multiple stays are required on a support, the minimum stay separation distance shall be 2.0m.

Multi-coloured stay guards shall be inserted on all stays that terminate into hedge lines to enhance visual identification.

Additional guidance on the installation of stays can be found in drawing 109119 3326 and OHI 4.

A balancing stay is required at transition positions between open wire and ABC constructions. See Appendix 10 for details. The stay shall be against the open wire line.

Struts, flying stays and outrigger stays may be used if necessary but should be avoided wherever possible. See drawing numbers 1091193331 & 1091193327 for additional information.

Alternative Stay Systems

Augured ground anchor stay systems and duckbill stay anchors may be used as alternative staying systems. See NSP/004/104 for additional guidance.

3.6 Erection Techniques

ABC conductor shall be erected using tension stringing techniques to ensure the conductor does not come into contact with the ground or abrasive surfaces. Prior to conductor sagging, conductors shall be pre-tensioned for 45mins with adjustment at 15 min intervals. The methods which shall be used, are fully described in guidance document NSP/004/105.

Tension joints are not allowed in new sections of line and conductors shall be erected as complete section lengths.

3.7 Materials

3.7.1 Conductor

The conductors shall comply with ENA TS 43-13 and shall be of compacted, circular, stranded aluminium having a black XLPE insulation. The completed cable is formed by twisting the four individual conductors together. (plus the 25mm protective conductor where required) The length of lay of the cores shall be 22-28 times the calculated diameter of the assembly as given in Appendix 2. Conductors shall be manufactured and supplied in accordance with material specification NPS/001/007.

Identification of Cores

The individual phase cores shall be identified as indicated on Drawing No. 1000431301 by 1, 2 or 3 longitudinally continuous raised ribs.

The relationship between the number of ribs and phases is to be as follows:

1 rib	-	red phase
2 ribs	-	yellow phase
3 ribs	-	blue phase

The surface of the neutral shall be lightly ribbed continuously along its length with a minimum of 16 ribs spaced evenly around the circumference of the core.

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Where an additional protective conductor is utilised it will be provided as a smooth surface with no ribbing.

Care must be taken when phasing in ABC networks with old historical open wire networks as the above standard phase sequence has not always been employed.

3.7.2 Wood Poles

Shall be treated and fabricated in accordance with the requirements of ENA TS 43-88 Parts 1 & 2 with water soluble based preservatives (non-creosote based) and supplied in accordance with material specification NPS/001/013.

Wood supports for new ABC networks shall be a minimum classification of Medium, in accordance with BS1990 and fabricated in accordance with drawing nos. 1091193301 sheet 4.

Drawing Nos. 1091193301 sheet 3 shows an alternative fabrication of a single M22 fixing hole located 300mm from the pole top for attachment of the ABC conductor and 6 additional holes at 150mm spacing to cater for historical 5 or 6 open wire networks.

3.7.3 Steel Poles

Galvanised tubular steel poles, minimum class 'K' manufactured in accordance with NPS/001/018 may also be utilised on unstayed angle and terminal positions where it is not possible to utilise conventional wood poles and stays. This option must only be used in special circumstances due the excessive cost and foundation requirements. Further details may be found in the renovation document NSP/004/041/001.

3.7.4 Wood poles - BT Joint Construction Agreement

Where it is desired to take advantage of the reciprocal arrangements for the attachment of wires to poles owned by British Telecom, the procedure to be adopted and the conditions which apply shall be as set out in Engineering Recommendation EB/BT2.

3.8 Materials and specific requirements for their application

3.8.1 General

All ABC fittings shall be in accordance with NPS/001/024 "Specification for Low Voltage Aerial Bundled Conductor Fittings" and as detailed in the assessed products database..

3.8.2 Insulation Piercing Connectors (IPC's)

IPC's shall be used to provide all connections onto the ABC system. The connectors shall be applied over the insulation of the conductor and tightened using the approved 13mm insulated spanner until shearing of the torque limiting device takes place. Rubber gloves and insulated tools must be worn and used when installing and/or disconnecting insulation piercing connectors from LV systems.

3.8.3 Service Distribution Boxes

This is a plastic weatherproof box for mounting on poles using stainless steel straps or direct fixing. It provides the facility for connecting up to 9 phase and neutral conductors. Single or three phase services will enter the box via the holes in the base and terminate into separate terminal blocks. Service distribution boxes are available to accommodate both 4 core PME networks and 5 core CEW networks

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3.8.4 Anchor Clamps

The ABC is terminated by means of a bolted type tension clamp, which grips all the insulated conductors between insulated mouldings placed either side of a central wedge. The anchor clamp is secured to the pole by means of an eyebolt. All anchor clamps shall, during assembly, be fitted in such a manner that only insulated components are in contact with the conductor insulation. Before clamping the ABC, the individual cores shall be adjusted to ensure that they enter in a uniform manner with no excess slack being exhibited by any conductor. The current range of clamps is range taking across all sizes of 4 core ABC.

An additional two anchor clamps are available for 2 core and 4 core 35mm² ABC to provide a smaller clamp for reduced tension service situations.

ABC bundles with a fifth core protective conductor shall be terminated using the standard 4 core anchor clamps as detailed above. To retain the protective core in its location within the main bundle, the protective core shall be tie wrapped to the main bundle prior to the bundle entering into the anchor clamp.

3.8.5 Roller / Suspension Clamps

The ABC is suspended at intermediate poles by means of a roller suspension clamp comprising a metallic clamp with a split neoprene sleeve and tightening bolt. When the suspension clamp is in the fully retracted position, the suspension clamp acts as a running out device.

All suspension clamps shall, during assembly, be fitted in such a manner that only insulated components are in contact with the conductor insulation. The maximum angle of line deviation imposed on a suspension clamp shall not exceed 60 degrees. A special extension roller assembly shall be fitted onto all suspension roller clamps where the angle of deviation exceeds 30 degrees. The extension rollers shall be removed once the conductor has been erected and terminated. Failure to use the extension rollers will result in the ABC jamming or snatching in the roller on heavy angles

Suspension clamps can be used at straight-line uplift positions where the vertical angle of deviation is less than 30 degrees. In situations where uplift conditions can occur suspension clamps shall be inverted and bolted into a fixed position or the pole position shall be sectioned.

The roller / suspension clamps are range taking across all sizes of ABC Bundle.

3.8.6 Weak Links

Weak link devices rated at 5kN may be incorporated in line suspension supports where the construction could be at risk from the effects of falling trees or branches. Weak links shall not be used on supports associated with crossings of carriageways, railways and navigable waterways.

Suspension angle supports will not incorporate weak links.

When a weak link has operated or it is suspected that an object has fallen onto the line, whether or not weak links are included in that section of line, all supports in that line section shall be inspected as follows:-

The attachment bolts of the anchor clamps shall be examined to determine whether they are deformed if bent they shall be replaced.

Intermediate supports should be examined for signs of the suspension clamp deforming due to excessive weight.

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Intermediate supports use M20 pigtail hook bolts and these should also be checked for signs of deformation

3.8.7 Wall Mounts

Wall mounts shall be used to secure the cable to walls in straight-line situations and they are to be installed at intervals of 1000 mm. Using this fitting permits the cable installation to be carried out after the wall mounts have been secured.

3.8.8 Wall Brackets

Wall brackets shall be used in conjunction with the appropriate rubber insert where deviations are necessary due to waste and rainwater pipes etc. and also where the cable is suspended between buildings.

3.8.9 Outriggers

The use of outrigger brackets is permitted where minimum clearances to buildings or structures cannot be achieved, or to avoid unacceptable angles of line deviation at unstayed pole positions **and** where there is no practicable alternative.

Outrigger brackets shall be arranged as shown on drawing Y602L0106 and shall be used at intermediate positions only. The use of outriggers shall be limited to line deviations up to 30°.

3.8.10 Pole Fixings

Shall be supplied in accordance with material specification NPS/001/011

Eyebolt and Eye nut

An M20 eyebolt is to be used in conjunction with an eye nut at in line section positions, to secure the anchor clamps on wood poles. It is to be installed so that the collar is pulled tightly against the pole before the clamps are finally located and therefore no sag adjustment is permitted by using the eyebolt and nut. Where drilled steel poles are used, eye nuts and eyebolts shall be used in the same manner as in wood poles.

Hook Bolt

Open pigtail hooks are to be used to secure suspension clamps to wood poles. A 22 mm diameter hole is required to accommodate this fitting.

Extended Hook Bolt

Provides additional clearance at suspension positions where cable is running through an outside angle.

3.8.11 Conductor Joints

Full tension joints

Due to the difficulties associated with balancing the tension across all four cores, full tension joints shall not be utilised on new ABC lines.

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Non tension Joints

Where non tension joints are created at section poles, the insulation of the system shall be restored through the use of an insulated sleeve.

3.9 Electrical Connections

All connections shall comply with the requirements of ENA TS 43-14 or 43-92 as appropriate and be supplied in accordance with material specification NPS/001/024

3.9.1 Main Line

The use of midspan tension fittings in an installation is not permitted. When construction joints are required they are to be positioned in the jumper loop at section supports.

Double bolt insulation piercing connectors to Drawing No 1000431414, sheet 2 are to be used for all mains cable branch connections, e.g. at tee connections.

Where the ABC bundle is cut at section positions for straight through connection purposes, non-tension connectors to Drawing No 1000439204 shall be used and covered with an insulating sleeve.

At interface with an open wire system, the ABC tails are to be connected to the open wire conductors using double bolt IPC's to Drawing No 1000431414 Sheet 3. (See arrangement drawing 1000431208).

Where the circuit design incorporates positions, which may be required for use as network isolation points, section fuses in accordance with drawing no. 1091471530 shall be used. Insulation piercing connectors are not to be used as disconnection/reconnection devices at such positions. See arrangement drawing 1000431207 for details.

Where existing single phase 3 wire systems are to be replaced, a four core ABC conductor shall be used. The yellow and blue phase conductors shall be connected together in parallel on the ABC tails at, or as near to, the source of supply. This may be achieved by utilising two double bolt insulation piercing connectors at each position. Care shall be taken to maintain the load balance across the phases.

Where the cut ends of an ABC bundle occur at the end of an LV network, the neutral conductor must be connected to earth. See the terminal support drawing 1000431209 for more information.

3.9.2 Service Connections

When hybrid service cables are used, both phase and neutral connections will be made using single bolt insulation piercing connectors to Drawing No 1000431414, sheet 1 ensuring where possible that a phase balance is maintained on the main. When the neutral connection is made, a wrap of four layers of PVC tape at 50 % overlap shall be applied to the stranded neutral prior to connection.

Code of practice NSP/004/043 and drawings 1090431214 provide additional detail on service connections.

A cable tie shall be applied to secure the tapping conductor to the conductor bundle otherwise work hardening of the service conductor may occur leading to eventual failure.

The application of electrical connections to the ABC shall not impair the environmental protection afforded by the insulation. Any connection applied following the removal of insulation shall be provided with an effective method of resealing the conductor against the effects of the environment and shall not reduce the mechanical and electrical strength of the insulation of the system. Due regard shall be taken of the possibility of insulation retraction.

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Where it is necessary to remove an insulation piercing connector, the conductor insulation shall be sealed against ingress of moisture by the use of a self amalgamating tape.

No insulation piercing connector shall be reapplied at the position of a previously removed connector.

Public lighting service cables shall be attached in a similar manner as the hybrid service neutral indicated above, with the addition that the conductors are doubled prior to the application of the PVC tape.

When the number of single phase CNE services is likely to exceed four, or three phase CNE services exceeds two, the service connections shall be obtained through the use of a service distribution box. The box is supplied by a short length of 4 x 35mm² ABC or 4 x 35mm² I and S cable using double bolt insulating piercing connectors to connect to the ABC main.

Where SNE concentric services are terminated into a service distribution box on CEW networks, the box shall be supplied by a short length of 5 x 35mm² I and S cable using double bolt insulating piercing connectors to connect to the ABC main plus protective conductor. The SNE earth wires shall be bunched together and terminated into the separate earth connector block located within the distribution box.

When a distribution box is fitted to a 4 core PME system then the neutral block must be connected to a separate earth electrode as detailed in clause 3.11 of this spec. Note where service boxes are used with CEW networks the earth block shall be earthed rather than the neutral conductor. See the additional earthing notes in clause 3.11.

3.9.3 Mains Cable Terminations

Underground cables on poles will be terminated using a heat shrink termination as in Drawing No 1000431210 and the ABC connected to it using non-tension compression joints. All connections being covered with insulation sleeve.

When terminating waveform cables the phase conductors shall be jointed using non-tension compression fittings, as detailed in the jointing Work Specification 02/364.

Consac cables shall be terminated in a similar manner to the above except the ABC neutral shall be connected to the neutral/earth bond bracket.

3.9.4 Transformer Poles

The LV arrangement on pole mounted transformers shall remain as shown on Drawing No 1091010130, the connection between the single core from the fuses and the ABC tails to be a bimetal compression fitting Drawing No 1091010649 which shall then be covered with an insulated sleeve.

3.9.5 Static Balancers and Regulators

Where ABC is erected at a balancer/regulator pole, it shall be connected using ABC tails for all inter connection between the auxiliary apparatus,

3.9.6 Conductor Ends

All cut ends of conductors will be insulated using approved cable end caps as specified in ENA TS 43-14 to provide a durable and effective means of insulation and sealing against ingress of moisture, with an insulation rating of 600/1000V.

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When the cut end forms the termination of an LV network the neutral conductor must be connected to earth as shown in Drawing No 1000431209.

3.10 Associated Materials

3.10.1 General

Clause 3.3.2 of this standard requires the use of additional non-conducting protection when contact by third parties with the ABC is possible from normal access positions; where there is a risk of mechanical damage and where in close proximity to metallic pipes or cables.

Details of positions where additional protection may be required are shown on Drawing No 431219.

3.10.2 Proximity Protection

Consists of a black non-conducting split tubing with a minimum wall thickness of 1.5mm to Drawing No. 431411. It shall be used where the ABC bundle is attached above 2.4m and there is deemed to be a potential risk of mechanical damage by the movement of vehicles or the placement of materials eg ladders against the bundle. The tubing shall be secured by cable ties ensuring that the bundle is totally enclosed.

3.10.3 Mechanical Protection

High impact black PVC cable guards shall be used in those situations where there is a risk of mechanical damage.

Where the ABC could be damaged by abrasion against some other object, protective measures shall be taken. For example, where ABC is cleated past a stay it shall be protected with rigid UPVC tubing as shown on drawing 1000431411.

3.11 Earthing

All new and refurbished overhead lines shall be designed to comply with IMP/010/011 "Code of practise for earthing LV Networks and HV Distribution Substations".

PME connections to the neutral conductor shall be made using a single double bolted Insulation Piercing Connector as shown on drawing 1000431414 sht2. The connection to earth shall be made using 32mm² Hard Drawn Copper insulated with Black PVC.

The earthing conductor shall be cleated directly to the pole using polypropylene cable cleats and terminated at an earth electrode buried 600mm below ground level. See drawing 1091193339 Sht1 & 2.

Where CEW systems are required to be retained this shall be replaced with an ABC bundle that includes the additional protective conductor. The protective conductors shall be connected to the CEW earth conductor at each end of the Overhead section.

Note

In accordance with IMP/010/011 clause 3.8.3, when CEW networks are reviewed and retained as a open wire bare system or converted to a 5 core ABC system "additional earth electrodes and neutral earth bond connections shall be installed at those positions at which earth electrodes would have been required if the network were being converted to PME"

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3.11.1 Guard wires for Over-running HV Conductors

Where ABC replaces open wire conductors and the conductors are beneath an HV line, the guard wire shall be retained in accordance with previous practice.

3.11.2 Earthing and bonding of steel poles

To counter the risk of a rise in potential in steel poles used on LV networks one of the two following actions should be taken.

a) Ensure that all steel poles are effectively earthed at the base of the pole in accordance with drawing no 1091193622 or the protective conductor of a CEW system.

b) Apply MEN to the overhead network by bonding the steel pole to the neutral conductor. This does not necessitate bringing the o/h network up to PME requirements merely requiring it to fulfil the principles of MEN networks.

Note.

MEN cannot be applied to networks with CEW (Continuous Earth Wires)

To convert an overhead network to MEN, an earth electrode shall be installed at the end of the network most remote from the substation to earth the neutral conductor. Once this has been achieved, adoption of MEN by bonding a steel pole to the neutral conductor is a satisfactory alternative to earthing the pole directly to an earth electrode"

If the network is re-conducted with ABC, the neutral bar and insulators associated with the open wire network shall be removed and an electrical connection made between the pole and the neutral conductor.

Where CEW networks are encountered on steel poles, the isolated neutral arrangement shall be retained. However the steel pole shall be bonded to the CEW wire.

3.11.3 Earthing arrangements for BT attachments

The earthing arrangements for BT attachments to company owned poles are set out in Engineering Recommendation PO5/1 "Protection of Telecommunication lines from Power Lines" and EB/BT2 "Conditions for BT and electricity suppliers - Joint use of poles"

3.12 Services

Overhead and underground services shall be arranged generally in accordance with drawing 100043121407. Where it is required to terminate either a single phase or a three phase ABC service at a wall bracket, this shall be done in accordance with drawing 1000431217 sht 1 or sht2. For additional information on services and service connections see NSP/004/043.

3.12.1 Service Disconnections from ABC

When it is necessary to disconnect a service at the pole top from the ABC network this shall be done by cutting and insulating the service tail outside of the IPC leaving sufficient length to allow subsequent reconnection.

Alternatively the insulation piercing connector may be removed and discarded. The insulation of the ABC shall be sealed against the ingress of moisture by the use of a self amalgamating tape and a wrap of one layer of PVC tape at 50% overlap.

Insulation piercing connectors shall not be reapplied in the position of a previously removed connector.

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3.13 Relationships with other bodies

Where lines to this standard are erected over or alongside the plant of Telecommunications Operators, Waterways Authorities or Network Rail operators, then the provisions of the relevant joint agreements shall apply. Details of the required clearances and any special provisions are available in Company Code of Practice NSP/004/011.

Additional information with respect to Telecommunication requirements can be found in EB/BT2 and PO5/1.

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4. References

4.1 External Documentation

Reference	Title
43-12	Insulated Aerial Bundled Conductor Erection Requirements for LV Overhead Distribution Systems
43-13	Aerial Bundled Conductors Insulated with Cross Linked Polyethylene for low voltage overhead lines.
43-14	Conductor Fittings and Associated Apparatus for use with Low Voltage Insulated Aerial Bare Conductors.
43-30	Low Voltage Overhead Lines On Wood Poles
43-88	Treatment of Wood Poles and associated Timber for overhead Lines
43-91	Stay Strands & Stay Fittings for Overhead Lines
43-92	Stay Strands & Stay Fittings for Overhead Lines
43-95	Steelwork for Overhead Lines
43-96	Fasteners and washers for Wood Pole Overhead Lines
BS 1990	Wood Poles for Overhead Power and Telecommunications Lines part 1
EB/BT2	Conditions for BT and electricity suppliers - Joint use of poles
PO5/1	Protection of Telecommunication lines from Power Lines

4.2 Internal documentation

Reference	Title
NSP/004/011	Guidance on Overhead Line Clearances
NSP/004/041/001	Specification for Renovation of existing Open Wire LV Lines to ABC or addition of conductors to open wire networks
NSP/004/043	Specification for Overhead Services, Surface Wiring and ABC Eves Mains
NPS/001/001	Technical Specification for Wood Poles and Associated Products for Overhead Lines
NPS/001/002	Technical Specification for Helical Products
NPS/001/005	Technical Specification for Steelwork, Conductor/Insulator Fittings and Stay Fittings for Overhead Lines and General Construction Works
NPS/001/007	Technical Specification for Conductors for Overhead Lines
NPS/001/010	Technical Specification for Notice Plates and Signs for Use on NEDL and YEDL Distribution Systems
NPS/001/011	Technical Specification for Fasteners And Fixings For Wood Pole Overhead Lines and General Construction Works
NPS/001/013	Technical Specification for Stay wire for Overhead Lines
NPS/001/016	Technical Specification for Compression Fittings and Mechanical Conductor Fittings for Overhead Lines
NPS/001/020	Technical Specification for Ground Anchor Systems for Use in Overhead Lines
NSP/004/104	Guidance notes on Stay types, make-off's and spreads
NSP/004/105	Guidance notes on conductor erection and dismantling
NSP/004/106	Guidance notes on conductor joints, terminations & binders
NSP/004/107	Guidance notes on conductor jumpers & non-tension connections
NSP/004/109	Guidance notes on anti climbing devices, signs and notices plates
IMP/010/011	Code of practise for earthing LV Networks and HV Distribution Substations

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4.3 Summary of Changes

Clause	Changes or Ammendments
2.2	Table of contents added
3.2.5	Additional note added to promote the availability of larger spans where the design of the LV network is not driven by the requirement for LV service take positions. Allowable spanning increased from 50m to 90m
3.4.1	Minor modification to pole sinking
3.4.2	Minor modifications to allowable unstayed angle poles and the use of augured poles foundations where site and ground conditions are suitable.
3.6	Conductor pre-tensioning time reduced from 60mins to 45mins
3.7.2	Further comments inserted to re-confirm that only non creosote treated pole are allowed on LV networks.
3.8.3	Note added to advise that service distribution boxes are now available for 5 wire CEW ABC networks
Drawing 1091193301	Drawing modified to show correct pole treatment and modified pole sinking depth

5. Definitions

Term	Definition
PME	Protective Multiple Earthing
MEN	Multiple Earthed Neutral
CEW	Continuous Earth Wire
Anchor Clamp	A fitting, which transfers the tensile forces within the tensioned conductor, cores to the supporting structure.
Suspension Clamp	A fitting that encloses and supports all ABC cores at an intermediate support.
Insulation Piercing Connectors (IPC's)	Connectors that can be applied to insulated conductors without the prior removal of the insulation
Main Lines	Lines of 4 core ABC with main cores of 50, 70, 95 or 120mm ² conductor
Service Lines	Lines of 2 or 4 core ABC with core sizes of 35mm ² conductor
Design Loading Conditions	These are the augmented ice and wind loadings corresponding to ice 9.5mm diameter, wind 380N/m ² , as specified in EATS 43-12
Terminal Support	A support where the ABC is attached by an anchor clamp and does not continue into an adjacent span.
Intermediate Support	A support where the ABC system is attached by a suspension clamp. Intermediate supports may have angles of deviation upto 60°.
Section Support	A support where adjacent spans are attached by means of anchor clamps. Section supports may have angles of deviation and may have provisions for electrically sectionalising the system. Unless there is a specific need for the conductors to be cut, the ABC shall be run unbroken and a loop, approximately 0.5m deep shall be provided.
I and S cable	Insulated & Sheathed Cable

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager		

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
A Bilclough	Overhead Programme Manager		

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
A Bilclough	Overhead Programme Manager		

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator		

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager		

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
M. Nicholson	Head of System Strategy		

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Technical Data Used As Design Basis For LV ABC - Appendix 1

ABC Size Related Data: ABC

General

Ultimate tensile strength (UTS) of aluminium 170 N/mm of nominal csa. The elastic limit of linear extensions and recovery assumed to be 60 per cent of UTS.

Modulus of elasticity (linear) 'E' 5600 hbar.

Coefficient of linear expansion 23.00×10^{-6} per degree centigrade.

The figures in brackets for 120 mm ABC are the calculated tension values based on the common 2.62 Ratio of Mass/m. Since the full strength characteristics of this conductor can never be fully utilised due to the limiting factors imposed by the pole strengths, the design loadings applicable to 4 x 95 mm ABC have been applied.

		Reduced Tension For Services		Full Tension					
ABC SIZES	sq mm	2 * 35	4 * 35	4 * 35	4 * 50	4 * 70	4 * 95	4 * 120	4 x 120 Plus 25
Bundle Diameter	mm	16.05	25.9	25.9	29.5	33.4	39.0	42.6	45.3*
Tension under Design Loadings	mwt			923 lb	1634 lb	2032 lb	2543 lb	2543 lb (2903 lb)	
	Kgf	87.8	132.6	418.1	743.40	922.0	1156.00	1156.00 (1317.00)	
	Kn	0.86	1.30	4.1	7.29	9.04	11.34	11.34 (12.92)	
Ratio of: UTS/Design Load		13.00	17.22	5.46	4.17	4.87	5.40	6.80 (5.97)	
Mass	Kg/m	0.26	0.52	0.52	0.70	0.96	1.30	1.60	1.70*
Ice Loaded Mass	Kg/m	0.59	1.06	1.06	1.16*	1.70	1.90 *	2.24*	2.38*
Wind Load on Iced ABC	Kg/m	0.99	1.37	1.37	1.51*	1.66	1.88 *	2.02*	2.12*
Ratio of Mass/m : Tension (10°C) x		13.3	13.3	4.15	2.62	2.62	2.62	(2.62)	(2.62)
Tension 10°C	Kgf	19.55	39.1	125.3	267.60	367.0	497.00	610.00	
	Kn	0.19	0.38	1.23	2.62	3.6	4.87	5.98	
Tension Limiting Device	Kn	-	-	-	4.0	4.0	4.0	4.0	

Note (**)

To extract the maximum benefits from the design sags, the common design ratio has been ignored. Instead all conductor sags have been derived from the MWT loadings.

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Extract From ENS TS 43-13 Issue 2 - Appendix 2

SCHEDULE OF TECHNICAL PARTICULARS FOR 600/1000 V AERIAL BUNDLED CONDUCTORS (ABC)

1	Nominal cross sectional area of conductors (mm ²)	35	35	50	70	95	120	120 + 25
2	Number of cores	2	4	4	4	4	4	4 x 120 1 x 25
3	Nominal number of wires in conductor	7	7	19*	19*	19*	19*	19*
4	Diameter of conductor: Minimum (mm) Maximum (mm)	6.6 7.5	6.6 7.5	7.7 8.5	9.3 10.2	11.0 12.0	12.0 13.2	12.0 13.2
5	Minimum average thickness of insulation (mm)	1.3	1.3	1.5	1.5	1.7	1.8	1.8
6	Minimum thickness of insulation at any point (mm)	1.07	1.07	1.25	1.25	1.43	1.52	1.52
7	Maximum diameter of core (excluding ribs) (mm)	10.7	10.7	12.1	13.8	16.1	17.6	17.6
8	Calculated maximum diameter of circumscribing circle of assembled bundle (mm)	21.9	26.2	29.6	33.7	39.3	42.6	45.2 (calc)
9	Approximate weight of bundle (kg/m)	0.26	0.52	0.70	0.96	1.3	1.6	1.70
10	Maximum dc resistance of conductor in bundle at 20°C (u/m)	868	868	641	443	320	253	253
11	Maximum ac resistance of conductor at 75°C (u/m)	1060	1060	783	542	392	310	310
12	Star reactance of bundle at 50 Hz (u/m)	86	86	83	80	80	76	76
13	Zero sequence impedance of bundle at 50 Hz and 75°C (u/m)	2120 +j86	4240 +j372	3130 +j348	2170 +j350	1570 +j348	1240 +j96	1240 +j96
14	Maximum continuous current carrying capacity per phase at 75°C (25°C ambient): For erection at pole top (A) For installation under eaves (A)	138 116	117 98	143 120	183 154	228 191	300 n/a	300 n/a
15	Maximum continuous conductor temperature (°C)	75	75	75	75	75	75	75
16	Ultimate tensile strength of conductor based on 170 N/mm ² (calculated) (kN)	5.6	5.6	7.6	11.0	15.3	19.3	n/a
17	Ultimate tensile strength of bundle (calculated) (kN)	11.3	22.4	30.4	44.0	61.2	77.2	n/a
18	Modules of Elasticity of Conductor (hbar)	5900	5900	5600	5600	5600	5600	n/a
19	Coefficient of Linear Expansion (1/°C)	23 x 10 ⁻⁶	23 x 10 ⁻⁶	23 x 10 ⁻⁶	23 x 10 ⁻⁶	23 x 10 ⁻⁶	23 x 10 ⁻⁶	n/a
20	Minimum bending radius of bundle (mm) Minimum bending radius of single core (mm)	175 86	210 86	236 98	270 110	312 129	341 141	341 141

* Subject to tolerance of 1 wire

Correction Factors for Maximum Continuous Current Ratings

AMBIENT TEMPERATURE °C	CORRECTION FACTOR
5	1.18
10	1.14
15	1.10
20	1.05

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Design Sag & Tension Data for 2 or 4 Core 35mm² (Extended service Spans) - Appendix 3

* ABC	35 sq.mm		0		TABLE 2 - DESIGN SAG AND TENSION							50m BASIC SPAN		*	

*Tension (N)	1419.80	1359.56	1311.50	1267.99	1228.38	1192.14	1158.82	1128.07	1099.58	1073.09	1048.37	1025.26	963.96	928.61	*
* (kgf)	144.78	138.64	133.74	129.30	125.26	121.56	118.17	115.03	112.13	109.42	106.90	104.55	98.30	94.69	*
* (lbf)	319.18	305.64	294.84	285.06	276.15	268.00	260.51	253.60	247.20	241.24	235.68	230.49	216.71	208.76	*
-----*															
* Temp. (C)	-5.60	.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00	65.00	75.00	*
-----*															
* Span	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	*
* (m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	*
-----*															
* 30	.40	.42	.44	.45	.47	.48	.50	.51	.52	.53	.55	.56	.60	.62	*
* 35	.55	.57	.60	.62	.64	.66	.67	.69	.71	.73	.74	.76	.81	.84	*
* 40	.72	.75	.78	.80	.83	.86	.88	.90	.93	.95	.97	.99	1.06	1.10	*
* 45	.91	.95	.98	1.02	1.05	1.08	1.11	1.14	1.17	1.20	1.23	1.26	1.34	1.39	*
* 50	1.12	1.17	1.22	1.26	1.30	1.34	1.38	1.41	1.45	1.49	1.52	1.55	1.65	1.72	*
* 55	1.36	1.42	1.47	1.52	1.57	1.62	1.66	1.71	1.75	1.80	1.84	1.88	2.00	2.08	*
* 60	1.62	1.69	1.75	1.81	1.87	1.92	1.98	2.03	2.09	2.14	2.19	2.24	2.38	2.47	*
* 65	1.90	1.98	2.05	2.12	2.19	2.26	2.32	2.39	2.45	2.51	2.57	2.63	2.79	2.90	*
* 70	2.20	2.30	2.38	2.46	2.54	2.62	2.70	2.77	2.84	2.91	2.98	3.05	3.24	3.36	*
* 75	2.53	2.64	2.73	2.83	2.92	3.01	3.09	3.18	3.26	3.34	3.42	3.50	3.72	3.86	*
* 80	2.87	3.00	3.11	3.22	3.32	3.42	3.52	3.62	3.71	3.80	3.89	3.98	4.23	4.39	*
* 85	3.24	3.39	3.51	3.63	3.75	3.86	3.97	4.08	4.19	4.29	4.39	4.49	4.78	4.96	*
* 90	3.64	3.80	3.94	4.07	4.20	4.33	4.46	4.58	4.70	4.81	4.92	5.04	5.36	5.56	*
* 95	4.05	4.23	4.39	4.54	4.68	4.83	4.96	5.10	5.23	5.36	5.49	5.61	5.97	6.20	*
* 100	4.49	4.69	4.86	5.03	5.19	5.35	5.50	5.65	5.80	5.94	6.08	6.22	6.61	6.86	*
-----*															
* BASIC DESIGN CRITERIA															

* U.T.S	22400 N				NOMINAL COND DIA		26.2 mm	PRE STRES		N/A Kg					
* F.O.S	5.463				COND. WT PER M		.52 Kg/M								

* M.W.T	4100 N				CO. EXP		.000023 /DEG C	DATE :							
* WIND LOAD	380 N/M				MOD. ELAS		59000 N/mm2	DGN NO.							
* ICE DIA	9.5 mm				BASIC SPAN		50 M								

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Design Sag & Tension Data for 50mm² - Appendix 4

* ABC	50 sq.mm		19/1.83mm		TABLE 2 - DESIGN SAG AND TENSION							50m BASIC SPAN			*		

*Tension (N)	3674.76	3277.45	2995.72	2765.96	2575.87	2416.32	2280.58	2163.68	2061.88	1972.35	1892.94	1821.94	1647.38	1554.68	*		
* (kgf)	374.72	334.21	305.48	282.05	262.67	246.40	232.56	220.63	210.25	201.12	193.03	185.79	167.99	158.53	*		
* (lbf)	826.12	736.80	673.47	621.81	579.08	543.21	512.70	486.41	463.53	443.40	425.55	409.59	370.35	349.51	*		

* Temp. (C)	-5.60	.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00	65.00	75.00	*		

* Span	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	*		
* (m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	*		

* 30	.21	.24	.26	.28	.30	.32	.34	.36	.37	.39	.41	.42	.47	.50	*		
* 35	.29	.32	.35	.38	.41	.44	.46	.49	.51	.53	.56	.58	.64	.68	*		
* 40	.37	.42	.46	.50	.53	.57	.60	.63	.67	.70	.73	.75	.83	.88	*		
* 45	.47	.53	.58	.63	.67	.72	.76	.80	.84	.88	.92	.95	1.05	1.12	*		
* 50	.58	.65	.72	.78	.83	.89	.94	.99	1.04	1.09	1.13	1.18	1.30	1.38	*		
* 55	.71	.79	.87	.94	1.01	1.07	1.14	1.20	1.26	1.32	1.37	1.42	1.58	1.67	*		
* 60	.84	.94	1.03	1.12	1.20	1.28	1.35	1.43	1.50	1.57	1.63	1.70	1.88	1.99	*		
* 65	.99	1.11	1.21	1.31	1.41	1.50	1.59	1.68	1.76	1.84	1.92	1.99	2.20	2.33	*		
* 70	1.14	1.28	1.40	1.52	1.63	1.74	1.84	1.94	2.04	2.13	2.22	2.31	2.55	2.70	*		
* 75	1.31	1.47	1.61	1.75	1.87	2.00	2.12	2.23	2.34	2.45	2.55	2.65	2.93	3.10	*		
* 80	1.49	1.68	1.83	1.99	2.13	2.27	2.41	2.54	2.66	2.78	2.90	3.01	3.33	3.53	*		
* 85	1.69	1.89	2.07	2.24	2.41	2.57	2.72	2.87	3.01	3.14	3.28	3.40	3.76	3.99	*		
* 90	1.89	2.12	2.32	2.51	2.70	2.88	3.05	3.21	3.37	3.52	3.67	3.81	4.22	4.47	*		
* 95	2.11	2.36	2.59	2.80	3.01	3.20	3.40	3.58	3.76	3.93	4.09	4.25	4.70	4.98	*		
* 100	2.34	2.62	2.86	3.10	3.33	3.55	3.76	3.97	4.16	4.35	4.53	4.71	5.21	5.52	*		

* BASIC DESIGN CRITERIA															*		

* U.T.S	30400	N	NOMINAL COND DIA				29.5	mm	PRE STRES			N/A Kg				*	
* F.O.S	4.17	COND. WT PER M				.7	Kg/M									*	

* M.W.T	7290	N	CO. EXP				.000023	/DEG C	DATE :							*	
* WIND LOAD	380	N/M	MOD. ELAS				56000	N/mm2	DGN NO.							*	
* ICE DIA	9.5	mm	BASIC SPAN				50	M									*

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Design Sag & Tension Data for 70mm² - Appendix 5

*****															*
* ABC	70 sq.mm		19/Comp		TABLE 2 - DESIGN SAG AND TENSION							50m BASIC SPAN			*
*****															*
*Tension (N)	5067.16	4513.38	4121.86	3803.28	3540.17	3319.66	3132.28	2971.05	2830.76	2707.46	2598.14	2500.44	2260.40	2133.01	*
* (kgf)	516.71	460.24	420.31	387.83	361.00	338.51	319.41	302.96	288.66	276.08	264.94	254.98	230.50	217.51	*
* (lbf)	1139.14	1014.65	926.63	855.01	795.86	746.29	704.17	667.92	636.38	608.66	584.08	562.12	508.16	479.52	*
-----															*
* Temp. (C)	-5.60	.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00	65.00	75.00	*
-----															*
* Span	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	*
* (m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	*
-----															*
* 30	.21	.23	.26	.28	.30	.32	.34	.36	.37	.39	.41	.42	.47	.50	*
* 35	.28	.32	.35	.38	.41	.43	.46	.49	.51	.53	.55	.58	.64	.68	*
* 40	.37	.42	.46	.50	.53	.57	.60	.63	.67	.70	.72	.75	.83	.88	*
* 45	.47	.53	.58	.63	.67	.72	.76	.80	.84	.88	.92	.95	1.05	1.12	*
* 50	.58	.65	.71	.77	.83	.89	.94	.99	1.04	1.09	1.13	1.18	1.30	1.38	*
* 55	.70	.79	.86	.94	1.01	1.07	1.14	1.20	1.26	1.31	1.37	1.42	1.57	1.67	*
* 60	.84	.94	1.03	1.11	1.20	1.28	1.35	1.43	1.50	1.56	1.63	1.69	1.87	1.99	*
* 65	.98	1.10	1.21	1.31	1.40	1.50	1.59	1.67	1.76	1.84	1.91	1.99	2.20	2.33	*
* 70	1.14	1.28	1.40	1.52	1.63	1.74	1.84	1.94	2.04	2.13	2.22	2.31	2.55	2.70	*
* 75	1.31	1.47	1.61	1.74	1.87	1.99	2.11	2.23	2.34	2.44	2.55	2.65	2.93	3.10	*
* 80	1.49	1.67	1.83	1.98	2.13	2.27	2.40	2.53	2.66	2.78	2.90	3.01	3.33	3.53	*
* 85	1.68	1.88	2.06	2.24	2.40	2.56	2.71	2.86	3.00	3.14	3.27	3.40	3.76	3.99	*
* 90	1.88	2.11	2.31	2.51	2.69	2.87	3.04	3.21	3.37	3.52	3.67	3.81	4.22	4.47	*
* 95	2.10	2.35	2.58	2.79	3.00	3.20	3.39	3.57	3.75	3.92	4.09	4.25	4.70	4.98	*
* 100	2.32	2.61	2.86	3.09	3.32	3.54	3.76	3.96	4.16	4.35	4.53	4.71	5.21	5.52	*
-----															*
* BASIC DESIGN CRITERIA															*
*****															*
* U.T.S	44000	N	NOMINAL COND DIA		33.7	mm	PRE STRES		N/A	Kg					*
* F.O.S	4.867		COND. WT PER M		.96	Kg/M									*
*****															*
* M.W.T	9040	N	CO. EXP		.000023	/DEG C	DATE :								*
* WIND LOAD	380	N/M	MOD. ELAS		56000	N/mm2	DGN NO.								*
* ICE DIA	9.5	mm	BASIC SPAN		50	M									*
*****															*
															*

Appendix 6

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Design Sag & Tension Data for 95mm² - Appendix 6

* ABC	95 sq.mm		19/2.52mm		TABLE 2 - DESIGN SAG AND TENSION							50m BASIC SPAN			*

*Tension (N)	6966.89	6193.60	5647.70	5204.36	4838.97	4533.33	4274.08	4051.37	3857.86	3688.01	3537.59	3403.30	3073.89	2899.38	*
* (kgf)	710.43	631.57	575.91	530.70	493.44	462.27	435.84	413.13	393.39	376.07	360.73	347.04	313.45	295.66	*
* (lbf)	1566.22	1392.38	1269.65	1169.99	1087.84	1019.13	960.85	910.78	867.28	829.10	795.28	765.09	691.04	651.81	*

* Temp. (C)	-5.60	.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00	65.00	75.00	*

* Span	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	*
* (m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	*

* 30	.21	.23	.25	.28	.30	.32	.34	.35	.37	.39	.41	.42	.47	.49	*
* 35	.28	.32	.35	.38	.40	.43	.46	.48	.51	.53	.55	.57	.64	.67	*
* 40	.37	.41	.45	.49	.53	.56	.60	.63	.66	.69	.72	.75	.83	.88	*
* 45	.46	.52	.57	.62	.67	.71	.76	.80	.84	.87	.91	.95	1.05	1.11	*
* 50	.57	.64	.71	.77	.82	.88	.93	.98	1.03	1.08	1.13	1.17	1.30	1.37	*
* 55	.69	.78	.85	.93	1.00	1.06	1.13	1.19	1.25	1.31	1.36	1.42	1.57	1.66	*
* 60	.82	.93	1.02	1.10	1.19	1.27	1.34	1.42	1.49	1.56	1.62	1.69	1.87	1.98	*
* 65	.97	1.09	1.19	1.29	1.39	1.49	1.58	1.66	1.75	1.83	1.90	1.98	2.19	2.32	*
* 70	1.12	1.26	1.38	1.50	1.61	1.72	1.83	1.93	2.02	2.12	2.21	2.29	2.54	2.69	*
* 75	1.29	1.45	1.59	1.72	1.85	1.98	2.10	2.21	2.32	2.43	2.53	2.63	2.92	3.09	*
* 80	1.46	1.65	1.81	1.96	2.11	2.25	2.39	2.52	2.64	2.77	2.88	3.00	3.32	3.52	*
* 85	1.65	1.86	2.04	2.21	2.38	2.54	2.69	2.84	2.98	3.12	3.25	3.38	3.75	3.97	*
* 90	1.85	2.08	2.29	2.48	2.67	2.85	3.02	3.19	3.35	3.50	3.65	3.79	4.20	4.45	*
* 95	2.06	2.32	2.55	2.76	2.97	3.17	3.36	3.55	3.73	3.90	4.07	4.23	4.68	4.96	*
* 100	2.29	2.57	2.82	3.06	3.29	3.52	3.73	3.93	4.13	4.32	4.50	4.68	5.18	5.50	*

* BASIC DESIGN CRITERIA															

* U.T.S	61200	N			NOMINAL COND DIA		39	mm	PRE STRES		N/A Kg				*
* F.O.S	5.396				COND. WT PER M		1.3	Kg/M							*

* M.W.T	11340	N			CO. EXP		.000023	/DEG C	DATE :						*
* WIND LOAD	380	N/M			MOD. ELAS		56000	N/mm2	DGN NO.						*
* ICE DIA	9.5	mm			BASIC SPAN		50	M							*

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Design Sag & Tension Data for 120mm² - Appendix 7

* ABC	120 sq.mm			Ø		TABLE 2 - DESIGN SAG AND TENSION						50m BASIC SPAN			*

*Tension (N)	6962.47	6363.62	5925.62	5557.46	5244.00	4973.91	4738.67	4531.81	4348.31	4184.29	4036.67	3902.97	3567.31	3385.04	*
* (kgf)	709.98	648.91	604.25	566.71	534.74	507.20	483.21	462.12	443.41	426.68	411.63	397.99	363.77	345.18	*
* (lbf)	1565.22	1430.60	1332.13	1249.37	1178.90	1118.18	1065.30	1018.79	977.54	940.67	907.48	877.42	801.96	760.99	*

* Temp. (C)	-5.60	.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00	65.00	75.00	*

* Span	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	Sag	*
* (m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	*

* 30	.25	.28	.30	.32	.34	.35	.37	.39	.41	.42	.44	.45	.49	.52	*
* 35	.35	.38	.41	.43	.46	.48	.51	.53	.55	.57	.60	.62	.67	.71	*
* 40	.45	.49	.53	.56	.60	.63	.66	.69	.72	.75	.78	.80	.88	.93	*
* 45	.57	.62	.67	.71	.76	.80	.84	.88	.91	.95	.98	1.02	1.11	1.17	*
* 50	.70	.77	.83	.88	.94	.99	1.03	1.08	1.13	1.17	1.21	1.26	1.37	1.45	*
* 55	.85	.93	1.00	1.07	1.13	1.19	1.25	1.31	1.36	1.42	1.47	1.52	1.66	1.75	*
* 60	1.01	1.11	1.19	1.27	1.35	1.42	1.49	1.56	1.62	1.69	1.75	1.81	1.98	2.09	*
* 65	1.19	1.30	1.40	1.49	1.58	1.67	1.75	1.83	1.91	1.98	2.05	2.12	2.32	2.45	*
* 70	1.38	1.51	1.62	1.73	1.83	1.93	2.03	2.12	2.21	2.30	2.38	2.46	2.69	2.84	*
* 75	1.58	1.73	1.86	1.99	2.10	2.22	2.33	2.43	2.54	2.64	2.73	2.83	3.09	3.26	*
* 80	1.80	1.97	2.12	2.26	2.39	2.52	2.65	2.77	2.89	3.00	3.11	3.22	3.52	3.71	*
* 85	2.04	2.23	2.39	2.55	2.70	2.85	2.99	3.13	3.26	3.39	3.51	3.63	3.97	4.19	*
* 90	2.28	2.50	2.68	2.86	3.03	3.19	3.35	3.51	3.65	3.80	3.94	4.07	4.45	4.69	*
* 95	2.54	2.78	2.99	3.19	3.38	3.56	3.74	3.91	4.07	4.23	4.39	4.54	4.96	5.23	*
* 100	2.82	3.08	3.31	3.53	3.74	3.94	4.14	4.33	4.51	4.69	4.86	5.03	5.50	5.79	*

* BASIC DESIGN CRITERIA															

* U.T.S	77200 N		NOMINAL COND DIA		42.6 mm		PRE STRES		N/A Kg						*
* F.O.S	6.807		COND. WT PER M		1.6 Kg/M										*

* M.W.T	11340 N		CO. EXP		.000023 /DEG C		DATE :								*
* WIND LOAD	380 N/M		MOD. ELAS		56000 N/mm2		DGN NO.								*
* ICE DIA	9.5 mm		BASIC SPAN		50 M										*

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Erection Data For Abc Attached To Buildings - Slack Spans @ reduced tensions - Appendix 8

SPAN METRES	35mm ² - SAG (m)	50mm ² - SAG (m)	70mm ² - SAG (m)	95mm ² - SAG (m)
10.0	0.16	0.19	0.24	0.26
12.5	0.26	0.30	0.38	0.41
15.0	0.37	0.43	0.54	0.58
17.5	0.50	0.58	0.74	0.80
20.0	0.66	0.76	0.97	1.04
22.5	0.83	0.96	1.23	1.32
25.0	1.03	1.19	1.51	1.62
27.5	1.24	1.44	1.83	1.96
30.0	1.48	1.71	2.18	2.34

NOTE:

Sags are based on design conditions applied to various sizes of ABC producing a tension not exceeding 1.3 kN.
The figures have been collated based on a 20m Basic Span @ 15°C

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Minimum Stay Spreads For Angle Poles - Appendix 9

50 mm² ABC

DEVIATION OVERALL LENGTH OF POLE (METRES)	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°	TERM
	MINIMUM STAY SPREAD IN METRES										
9.0	1.1	1.3	1.5	1.7	1.8	2.1	2.2	2.5	2.6	2.8	2.3
9.5	1.2	1.4	1.6	1.8	1.9	2.2	2.4	2.6	2.8	3.0	2.2
10.0	1.3	1.4	1.7	1.9	2.2	2.3	2.5	2.8	3.0	3.1	2.5
10.5	1.5	1.7	2.0	2.2	2.5	2.7	3.0	3.2	3.5	3.7	3.0
11.0	1.8	2.0	2.3	2.6	2.8	3.2	3.3	3.7	3.9	4.1	3.3
11.5	1.9	2.2	2.6	3.0	3.1	3.5	3.9	4.1	4.5	4.7	3.7
12.0	2.2	2.5	2.9	3.3	3.7	3.9	4.3	4.5	5.0	5.2	4.3
13.0	2.4	2.8	3.2	3.6	3.9	4.3	4.7	5.0	5.5	5.7	4.5

70 mm² ABC

DEVIATION OVERALL LENGTH OF POLE (METRES)	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°	TERM
	MINIMUM STAY SPREAD IN METRES										
9.0	1.1	1.3	1.5	1.8	1.9	2.1	2.3	3.1	3.2	2.9	2.3
9.5	1.4	1.5	1.8	2.1	2.2	2.5	2.6	3.3	3.4	3.3	2.8
10.0	1.6	1.7	2.0	2.3	2.7	2.8	3.1	3.5	3.6	3.8	3.1
10.5	1.8	2.2	2.3	2.7	3.0	3.3	3.7	3.9	4.2	4.4	3.7
11.0	2.0	2.5	2.8	3.2	3.3	3.7	4.3	4.5	4.7	5.1	4.3
11.5	2.2	2.8	3.1	3.5	3.9	4.3	4.9	4.9	5.4	5.8	4.7
12.0	2.5	3.1	3.5	3.9	4.3	4.8	5.9	5.6	6.1	6.4	5.4
13.0	2.8	3.2	3.9	4.3	4.7	5.2	5.7	6.2	6.5	7.0	5.7

95/120 mm² ABC

DEVIATION OVERALL LENGTH OF POLE (METRES)	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°	TERM
	MINIMUM STAY SPREAD IN METRES										
9.0	1.5	1.8	2.2	2.5	2.8	3.1	3.5	3.8	4.2	4.5	3.8
9.5	1.6	1.9	2.4	2.6	2.8	3.3	3.8	4.1	4.4	5.0	3.4
10.0	1.9	2.2	2.5	2.8	3.1	3.6	4.0	4.4	4.7	5.3	4.0
10.5	2.2	2.5	3.0	3.3	3.7	4.1	4.4	4.8	5.2	5.6	4.6
11.0	2.5	2.8	3.3	3.7	4.3	4.7	5.1	5.5	6.0	6.2	5.3
11.5	2.8	3.3	3.7	4.3	4.7	5.2	5.8	6.3	6.5	7.0	6.1
12.0	3.1	3.7	4.3	4.8	5.4	5.9	6.4	6.9	7.4	8.0	6.6
13.0	3.4	3.9	4.5	5.2	5.7	6.5	7.0	7.5	8.1	8.7	7.3

NOTES:

1. Preferred stay spreads are 45° as shown in Chart 4 but where these are not attainable then the above table may be used.
2. These minimum stay spreads relate to medium poles and single 7/4.00 mm stay.
3. The loadings are resultant forces derived from vertical and horizontal forces, using a wind span of 90 metres.
4. Calculations are based on UTS of poles and staywire compared with design loadings x 2.5.

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Out Of Balance Stays At In Line Junction Between Open Wire (O/W) And ABC Systems Are To Be As Follows: - Appendix 10

OVERALL LENGTH OF POLES	TYPE OF JUNCTIONS AND MINIMUM STAY SPREADS (M)											
	GROUP 1			GROUP 2			GROUP 3			GROUP 4		
	50/95/120mm ABC AND 4 x 32 Cu OR 4 x 50 A1 O/W			50/95/120mm ABC AND 4 x 32 Cu OR 4 x 50 A1 O/W + S/W WIRE			50/70/95/120mm ABC AND 4 x 70 Cu OR 4 x 100 Cu OR 4 x 100 A1			50/70/95/120mm ABC AND 4 x 70 Cu OR 4 x 100 Cu OR 4 x 100 A1 O/W + S/W WIRE		
	50mm ²	70mm ²	95/120mm ²	50mm ²	70mm ²	95/120mm ²	50mm ²	70mm ²	95/120mm ²	50mm ²	70mm ²	95/120mm ²
	ABC	ABC	ABC	ABC	ABC	ABC	ABC	ABC	ABC	ABC	ABC	ABC
9.0m	1 @ 3.5m	1 @ 2.8m	1 @ 2.1m	2 @ 4.0m	1 @ 4.7m	1 @ 3.7m	2 @ 4.3m	2 @ 4.0m	1 @ 4.3m	2 @ 3.8m*	2 @ 3.5m*	2 @ 3.1m*
9.5m	1 @ 3.8m	1 @ 3.0m	1 @ 2.2m	2 @ 4.6m	1 @ 5.0m	1 @ 3.9m	2 @ 6.0m	2 @ 4.6m	1 @ 4.6m	2 @ 4.1m*	2 @ 3.8m*	2 @ 3.3m*
10.0m	1 @ 4.0m	1 @ 3.1m	1 @ 2.3m	2 @ 5.3m	1 @ 5.3m	1 @ 4.2m	2 @ 6.8m	2 @ 5.3m	1 @ 4.9m	2 @ 4.4m*	2 @ 4.0m*	2 @ 3.5m*
10.5m	1 @ 4.2m	1 @ 3.7m	1 @ 2.7m	2 @ 6.1m	1 @ 5.6m	1 @ 4.6m	2 @ 6.8m	2 @ 6.1m	1 @ 5.2m	2 @ 4.6m*	2 @ 4.2m*	2 @ 3.7m*
11.0m	1 @ 4.9m	1 @ 4.1m	1 @ 3.0m	2 @ 6.9m	1 @ 6.2m	1 @ 5.1m	2 @ 7.7m	2 @ 6.9m	1 @ 6.0m	2 @ 4.9m*	2 @ 4.5m*	2 @ 3.9m*

NOTES:

- Preferred stay spreads are 45° as shown in Chart 4 but where these are not attainable the above table may be used.
- These stay spreads relate to medium poles with 1 x 7/4.00 stay or 2 x 7/4.00 stay strands (grade 700) as specified. (Spreads ending in * are Stout grade poles)
- These stays are to be installed against the open wire network.
- Where steel poles (historic type D or present day type K) are used with the above conductors at each end of the ABC span no stays are required.
- For situations other than the above refer to the Overhead Line Standards & Approvals Manager.
- Calculations based on 45, spans.
- Calculations are based on UTS of poles and staywire compared with design loadings x 2.5

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Drawings Index - **Appendix 11**

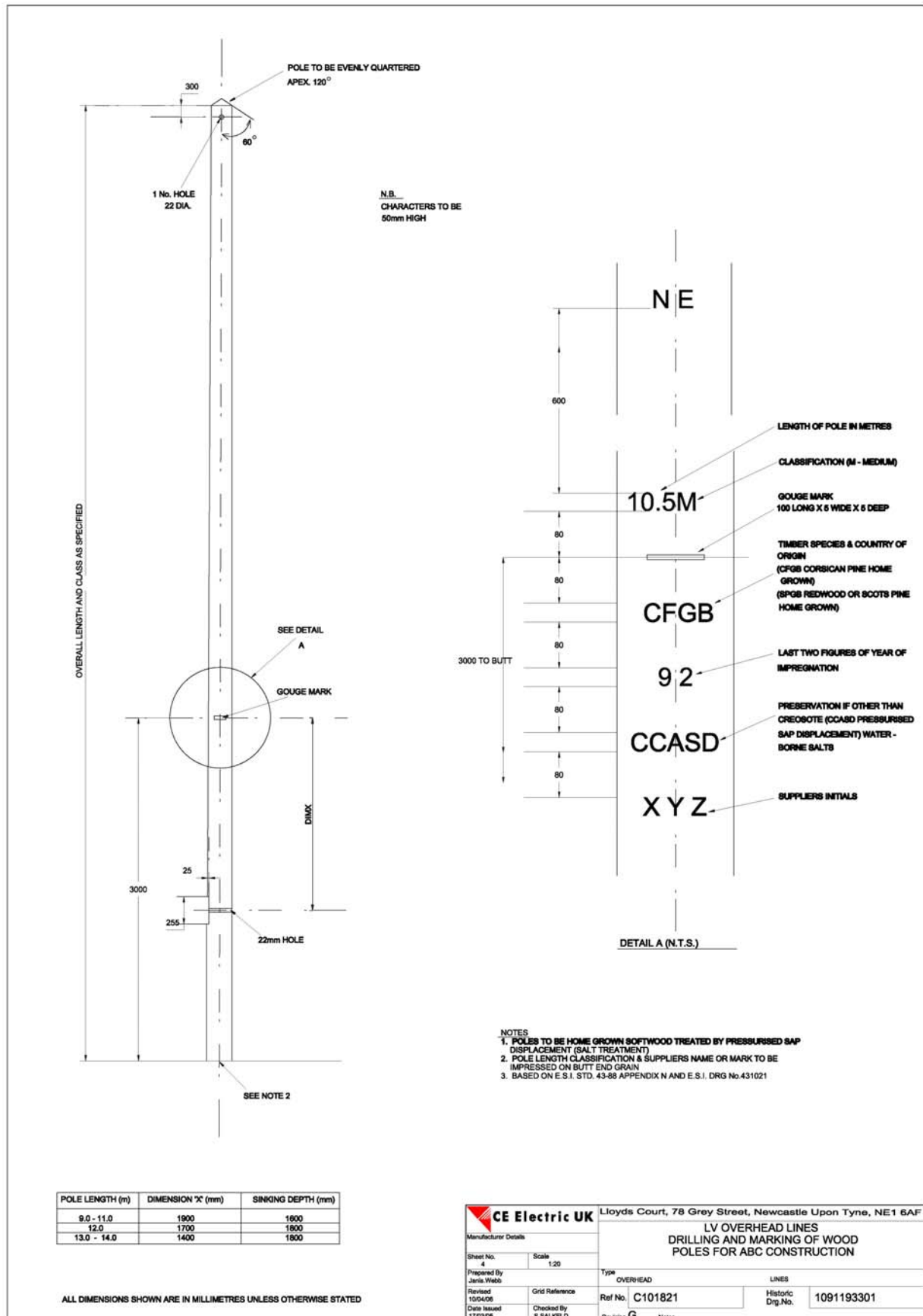
The following are typical drawings.

TITLE	DRAWING NO.
Drilling and Marking of Wood Poles for LV Construction using ABC	1.09.119.3301 Sh 4
ABC Fitting - Hook Bolt - Extended	1.00.043.1401
ABC Fitting - Hook Nut	1.00.043.1401 sht 2
ABC Fitting - Hook Bolt	1.00.043.1402
ABC Fitting - Wall Mount Hook	1.00.043.1404
ABC Fitting - Weak Link	1.00.043.1405
ABC Fitting - Suspension Clamp / Roller	1.00.043.1406 sht 2
ABC Fitting - Suspension Clamp / Roller - Extension Roller	1.00.043.1406 sht 3
ABC Fitting - Wall Mount Bracket	1.00.043.1407
ABC Fitting - Wall Mount Mains Support	1.00.043.1409
ABC Fitting - Anchor Clamp	1.00.043.1410
ABC Fitting - Wall Mounting Protective Sleeve	1.00.043.1411
ABC Fitting - Conductor End Caps	1.00.043.1412
ABC Fitting - Bimetal Insulation Piercing Connectors	1.00.043.1414 sh 1,2&3
ABC Fitting - Service Distribution Box	1.00.043.1416
Cable Pull Stocking	1.09.145.0015
ABC Come Along Clamp	1.09.145.0016
ABC Core Separator Wedge	1.09.145.0017
Unstayed Intermediate Support	1.00.043.1202
Intermediate Support up to 30° Line Deviation	1.00.043.1203
Intermediate Inside Angle Support up to 30 ° Line Deviation	1.00.043.1204
Intermediate Inside Angle Support up to 60 ° Line Deviation	1.00.043.1205
Section Support, Angles 20° -90° Line Deviation	1.00.043.1206
Section Support with Circuit Disconnecter	1.00.043.1207
Transition Support ABC. To Open Wire System	1.00.043.1208
Terminal Support	1.00.043.1209
Terminal Support, with Underground Cable	1.00.043.1210
Tee off from Intermediate Support	1.00.043.1211
Tee off Section Support	1.00.043.1212
Multiple Service Distribution Box	1.00.043.1213
Single Phase Service from Intermediate Support	1.00.043.1214
Multiple Service Take Off	1.00.043.1215
Intermediate Angle Support Without Strut or Stay	1.00.043.1216
Typical Attachments on and between Buildings	1.00.043.1217 Sh 1
Typical Attachments to Buildings	1.00.043.1217 Sh 2
Areas Requiring Proximity Protection	1.00.043.1219
Minimum Clearances of Aerial Systems to Buildings and Structures	1.00.043.1220
Static Balancer on Wood Pole	1.00.043.1222
Three Phase Regular on Wood Pole	1.00.043.1223
Sequence and Dimensions of Phase Identification Ribs on Cores of ABC Cable	1.00.043.1301

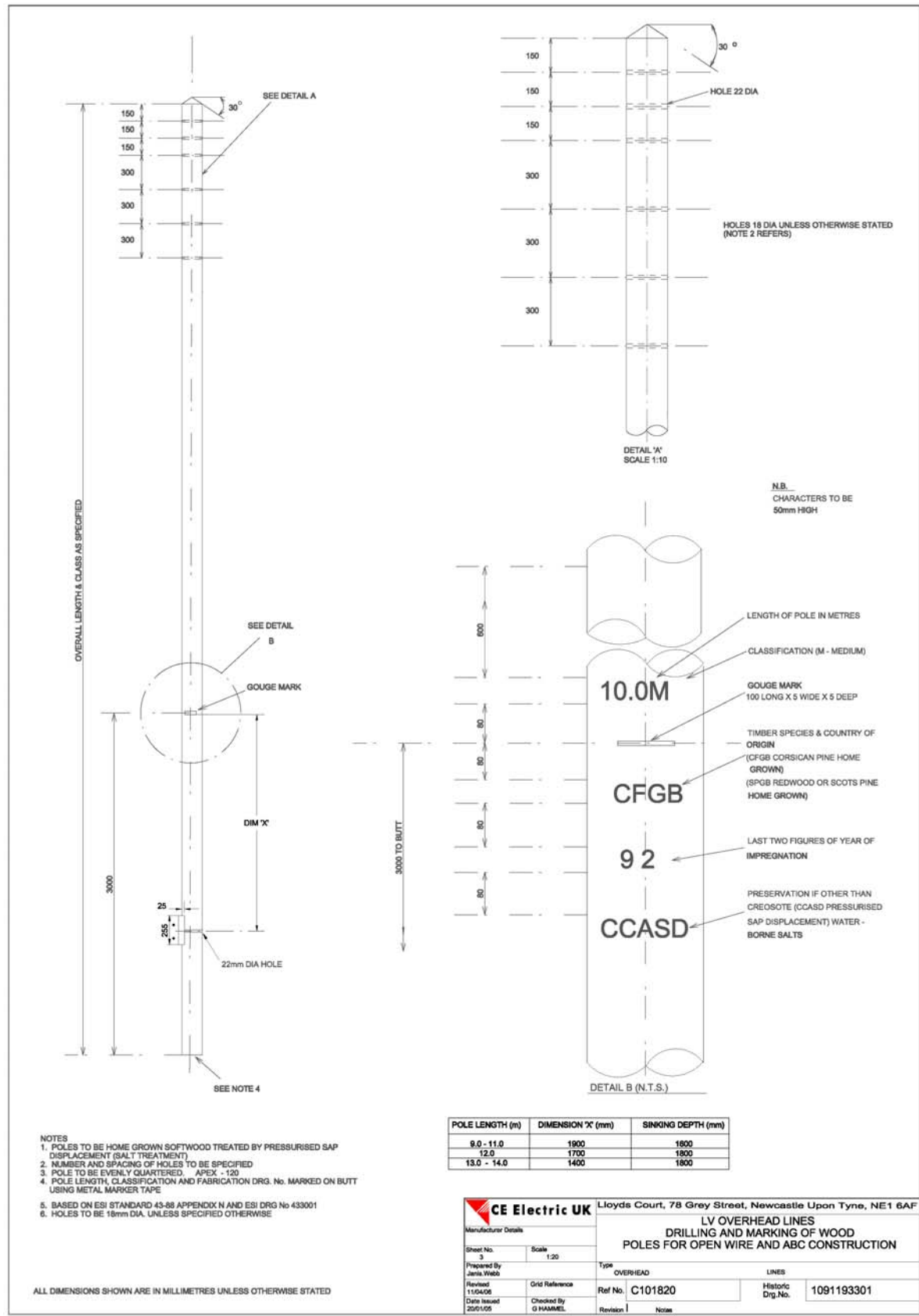
Document reference		NSP/004/041					
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Arrangement Drawings and Materials Lists - Appendix 12

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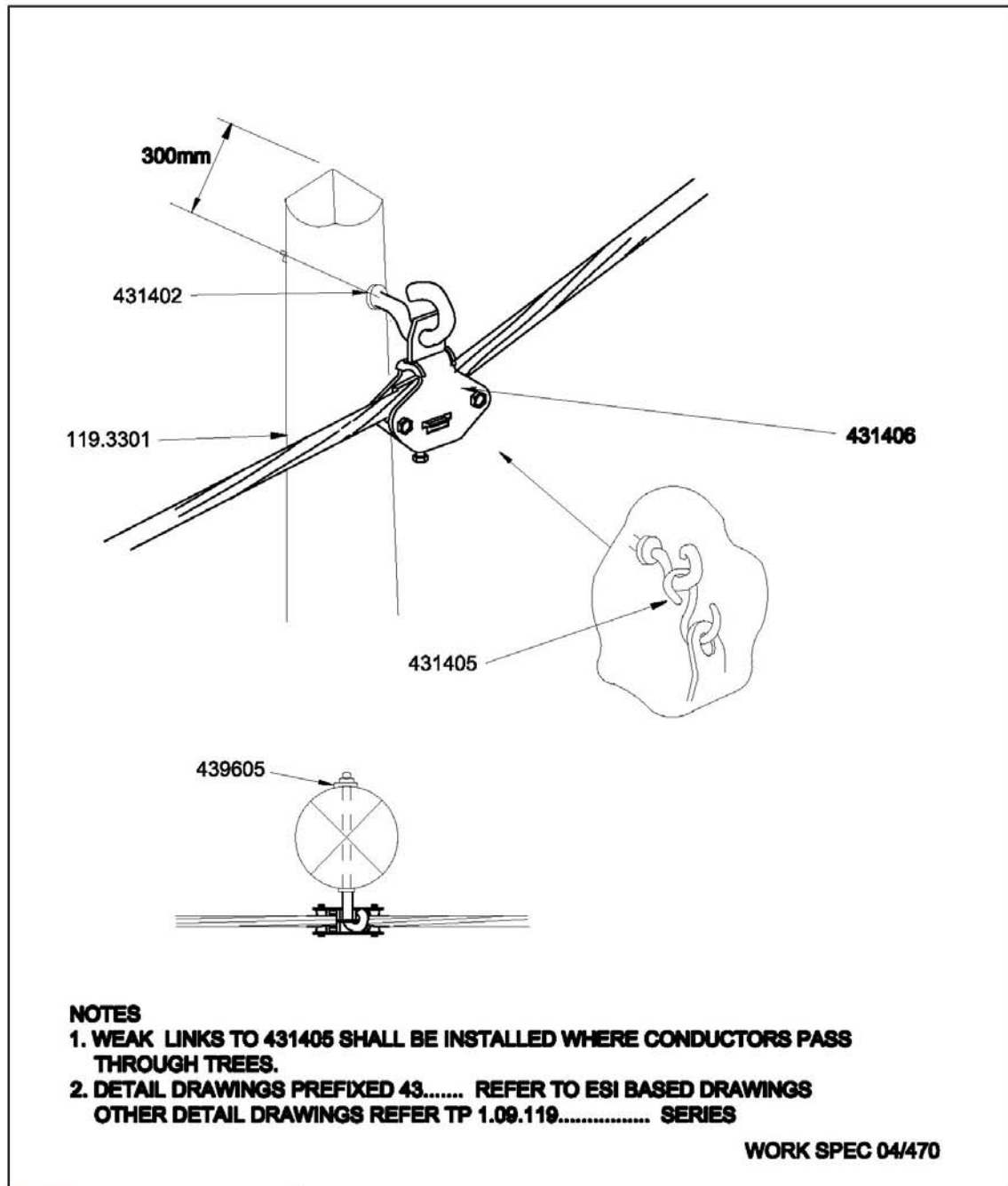
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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
		UNSTAYED INTERMEDIATE SUPPORT			
Manufacturer Details					
Sheet No. 1	Scale N.T.S.				
Prepared By Jonis.Webb		Type ESI		ESI	
Revised 25/04/06	Grid Reference	Ref No.	C316559	Historic Drg.No.	1000431202
Date Issued 23/09/05	Checked By G.HAMMEL	Revision	B	Notes	

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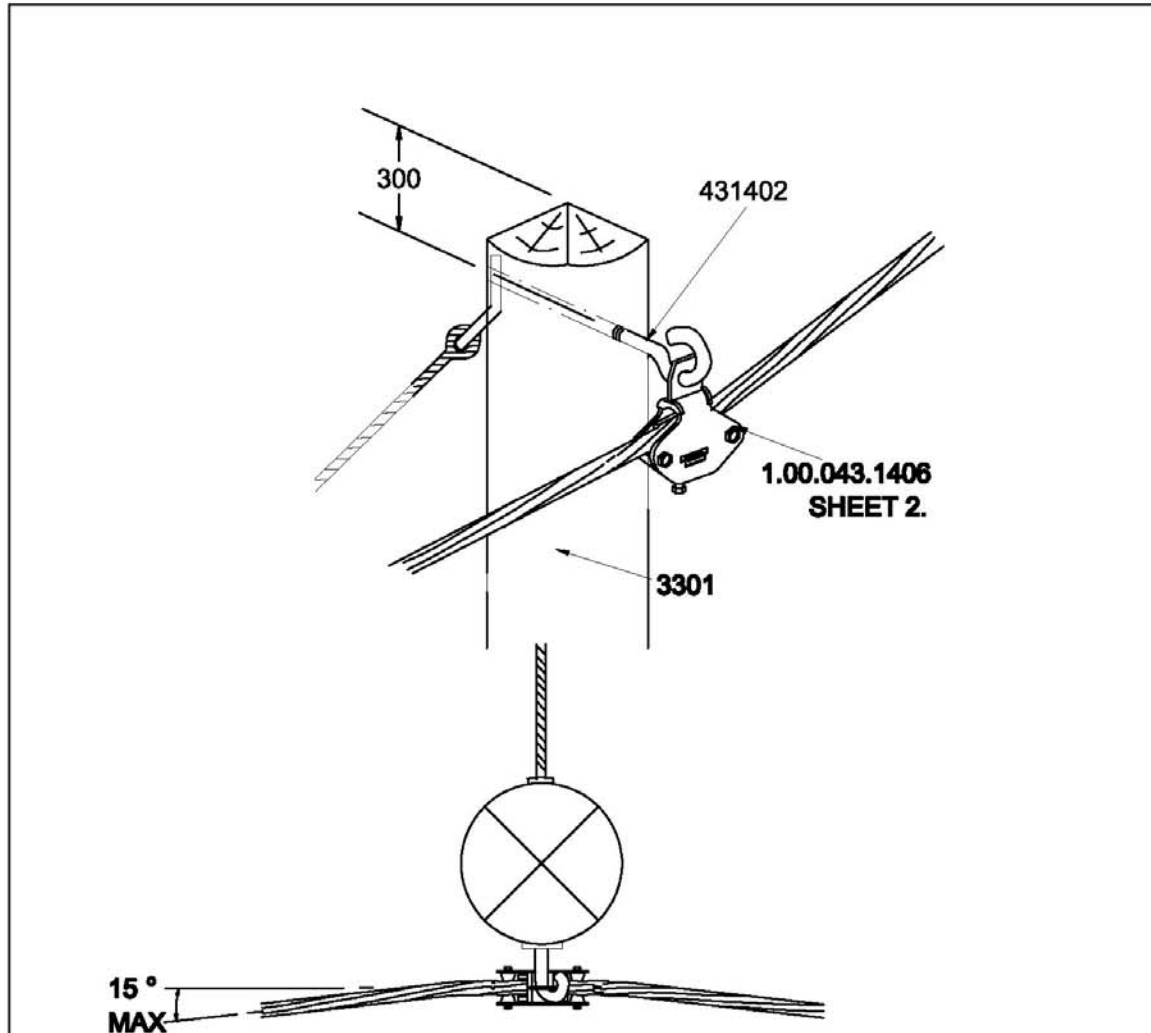
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220 mm	431402	1	265596
Washer, Square/Curved	434605	1	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Signs & Notices			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
When Specified:			
Mechanical Weak Link	431405	1	262032

Arrangement of Typical ABC Unstayed Intermediate Support

Drawing Number 1000431202

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NOTES:-

1. FOR ANGLES OF DEVIATION GREATER THAN 30°
2. DETAIL DRAWINGS PREFIXED 43..... REFER TO ESI BASED DRAWINGS AND OTHER DETAIL DRAWINGS REFER TO DRG 1.09.119..... SERIES

WORK SPEC No 04/471

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT OF TYPICAL ABC ANGLE SUPPORT UP TO 30* LINE DEVIATION			
Sheet No. 1	Scale N.T.S.	Type ESI			
Prepared By Janis.Webbb		ESI			
Revised 25/04/06	Grid Reference	Ref No.	C147597	Historic Drg.No.	1000431203
Date Issued 23/09/05	Checked By G HAMMEL	Revision D	Notes		

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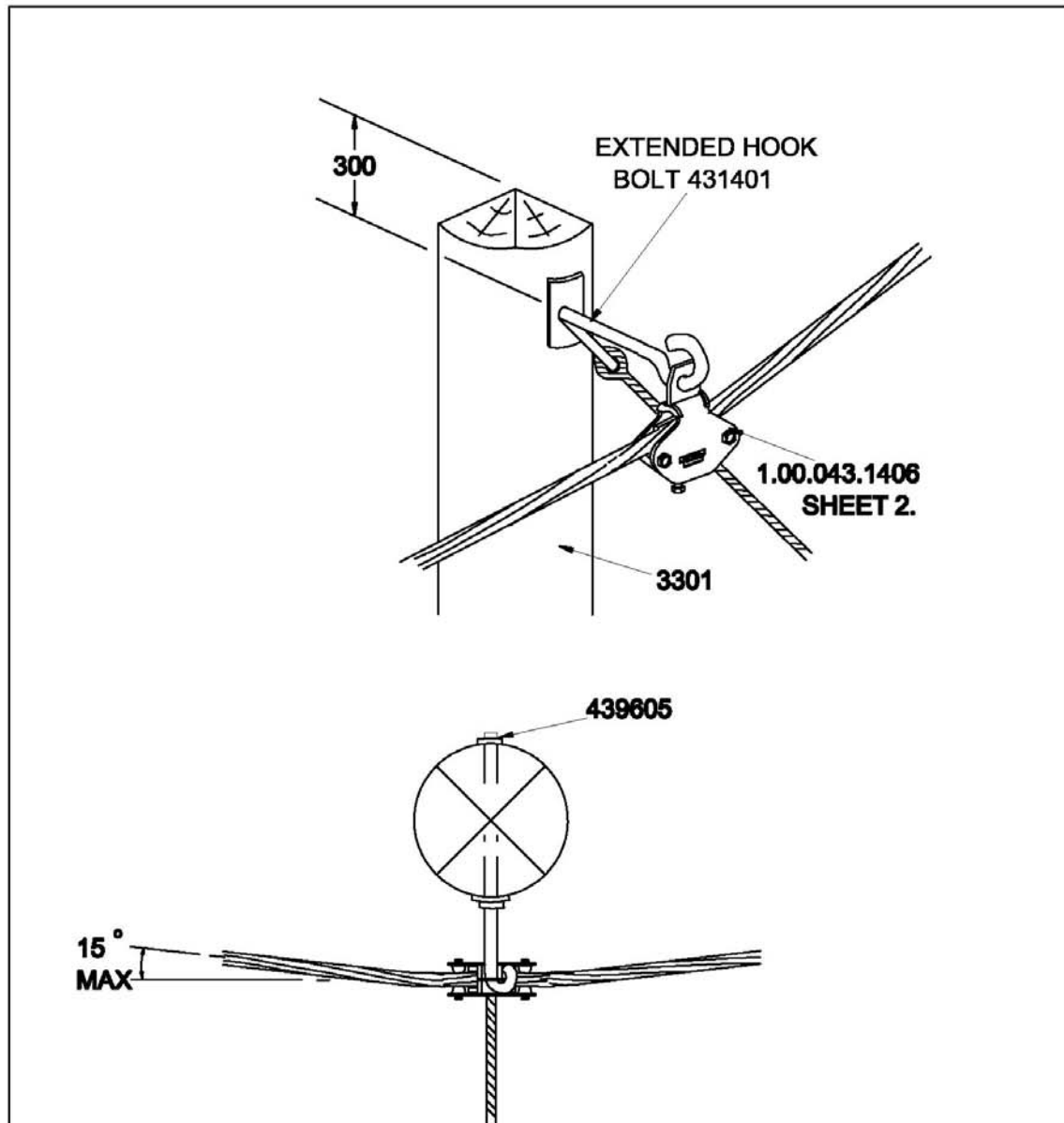
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220 mm	431402	1	265596
Washer, Square/Curved	434605	1	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
<i>Stay Components (Were Required)</i>			
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124
When Specified:			
Mechanical Weak Link	431405	1	262032

Arrangement of Typical ABC Angle support Up to 30° Deviation

Drawing Number 1000431203

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NOTES:-

1. DETAIL DRAWINGS PREFIXED 43.... REFER TO E.S.I.BASED DRAWINGS AND OTHER DETAIL DRAWINGS REFER TO 1.09.119.....SERIES.

WORK SPEC.No. 04/472

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
		ARRANGEMENT FOR TYPICAL ABC INSIDE ANGLE SUPPORT UP TO 30* LINE DEVIATION			
Manufacturer Details					
Sheet No.	Scale				
1	N.T.S.				
Prepared By Jonis.Webb		Type	ESI	ESI	
Revised 25/04/06	Grid Reference	Ref No.	C147598	Historic Org.No.	1000431204
Date Issued 23/09/05	Checked By G HAMMEL	Revision	E	Notes	

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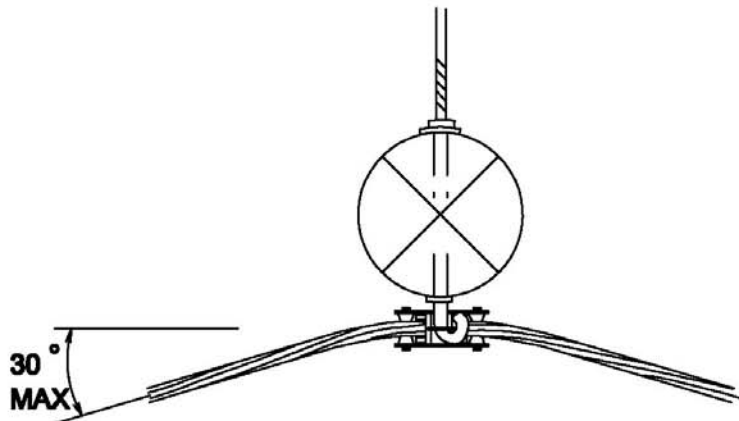
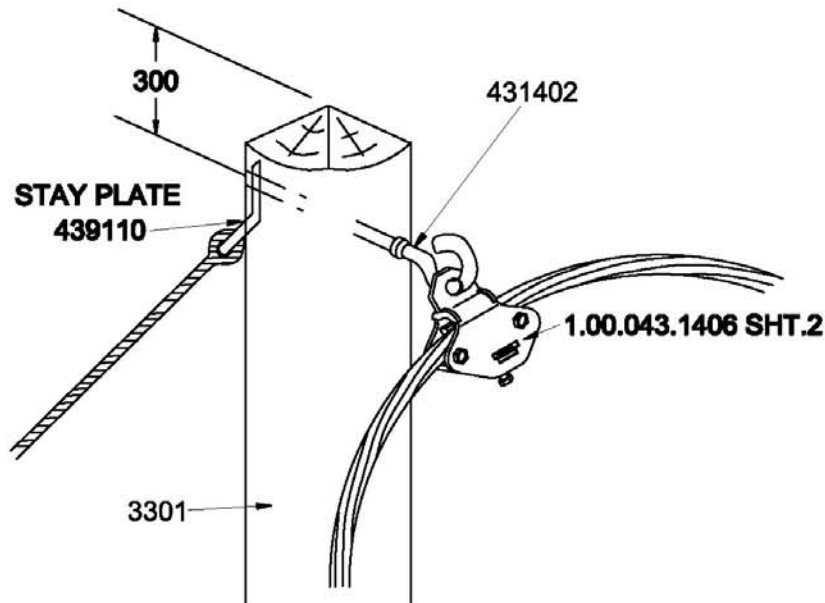
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, Extended M20, 240 mm	431401	1	265613
Washer, Square/Curved	434605	1	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
<i>Stay Components (Were Required)</i>			
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124
When Specified:			
Mechanical Weak Link	431405	1	262032

Arrangement of Typical ABC Inside Angle support Up to 30° Deviation

Drawing Number 1000431204

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NOTES:-

1. FOR ANGLES OF DEVIATION GREATER THAN 60° THE CONDUCTORS MUST **BE SECTIONED SEE DRG.1.00.043.1208**
2. DETAIL DRAWINGS PREFIXED 43.... REFER TO E.S.I.BASED DRAWINGS AND OTHER DETAIL DRAWINGS REFER TO 1.09.119.....SERIES.

WORK SPEC.No. 04/473

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC IINTERMEDIATE SUPPORT UP TO 60 DEGREES LINE DEVIATION			
Sheet No. 1	Scale	Type ESI			
Prepared By Barbara.Simpson					
Revised 18/05/06	Grid Reference	Ref No.	C406251	Historic Org.No.	1000431205
Date Issued	Checked By G HAMMEL	Revising	E	Notes	

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MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 240 mm	431402	1	265596
Washer, Square/Curved	434605	1	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
<i>Stay Components (Were Required)</i>			
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124
When Specified:			
Mechanical Weak Link	431405	1	262032

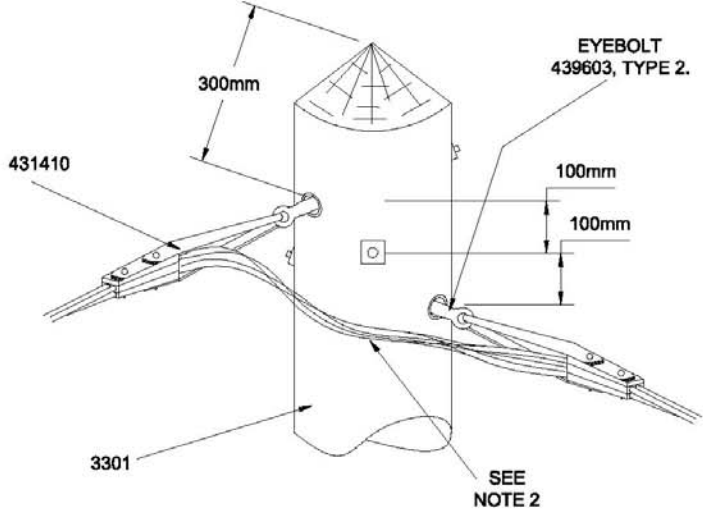
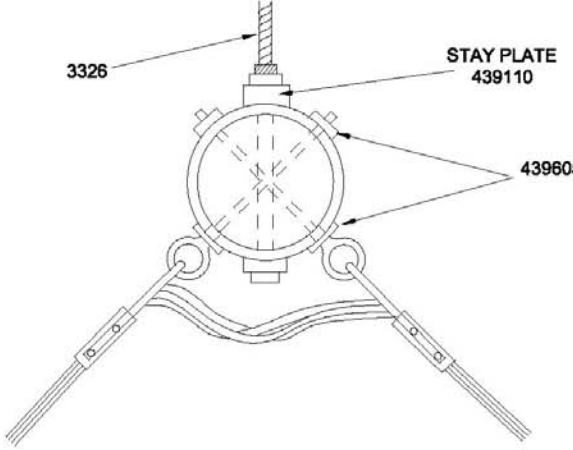
Arrangement of Typical ABC Angle support Up to 60° Deviation

Drawing Number 1000431205

Note

- Where angles of deviation exceed 30° the combined angle roller suspension clamp must be supplemented with extension rollers during the conductor running out stage. See drawing 1.00.043.1406 sheet 3 for details on the additional rollers.

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 SEPARATION OF CROSS DRILLING TO BE A MINIMUM DISTANCE OF 100mm BETWEEN CENTRES																													
																													
NOTES 1. DETAIL DRAWINGS PREFIXED 43..... REFER TO ESI BASED DRAWINGS AND OTHER DETAIL DRAWINGS REFER TO 1.09.119 SERIES 2. WHERE THE BUNDLE IS CUT AT THIS POINT NON TENSION CONNECTORS TO DRG No 43.9204 SHALL BE USED COVERED WITH AN INSULATING SLEEVE (CAT No 262210) 3. WHERE THE ANGLE OF DEVIATION IS LESS THAN 20° USE THE ARRANGEMENT SHOWN ON DRAWING No 43.1207																													
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="2" style="text-align: center;">  CE Electric UK </td> <td colspan="2" style="text-align: center;"> Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF </td> </tr> <tr> <td colspan="2" style="text-align: center;"> ARRANGEMENT FOR TYPICAL ABC SECTION SUPPORT ANGLE 20° - 90° LINE DEVIATION </td> <td colspan="2"></td> </tr> <tr> <td colspan="2" style="text-align: center;"> Manufacturer Details </td> <td colspan="2"></td> </tr> <tr> <td style="text-align: center;"> Sheet No. 1 </td> <td style="text-align: center;"> Scale </td> <td colspan="2"></td> </tr> <tr> <td colspan="2" style="text-align: center;"> Prepared By Barbara.Simpson </td> <td colspan="2" style="text-align: center;"> Type ESI </td> </tr> <tr> <td style="text-align: center;"> Revised 18/05/08 </td> <td style="text-align: center;"> Grid Reference </td> <td style="text-align: center;"> Ref No. C147599 </td> <td style="text-align: center;"> Historic Drg.No. 1000431206 </td> </tr> <tr> <td style="text-align: center;"> Date Issued 23/09/05 </td> <td style="text-align: center;"> Checked By G HAMMEL </td> <td colspan="2" style="text-align: center;"> Revision E Notes </td> </tr> </table>		 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		ARRANGEMENT FOR TYPICAL ABC SECTION SUPPORT ANGLE 20° - 90° LINE DEVIATION				Manufacturer Details				Sheet No. 1	Scale			Prepared By Barbara.Simpson		Type ESI		Revised 18/05/08	Grid Reference	Ref No. C147599	Historic Drg.No. 1000431206	Date Issued 23/09/05	Checked By G HAMMEL	Revision E Notes	
 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF																											
ARRANGEMENT FOR TYPICAL ABC SECTION SUPPORT ANGLE 20° - 90° LINE DEVIATION																													
Manufacturer Details																													
Sheet No. 1	Scale																												
Prepared By Barbara.Simpson		Type ESI																											
Revised 18/05/08	Grid Reference	Ref No. C147599	Historic Drg.No. 1000431206																										
Date Issued 23/09/05	Checked By G HAMMEL	Revision E Notes																											

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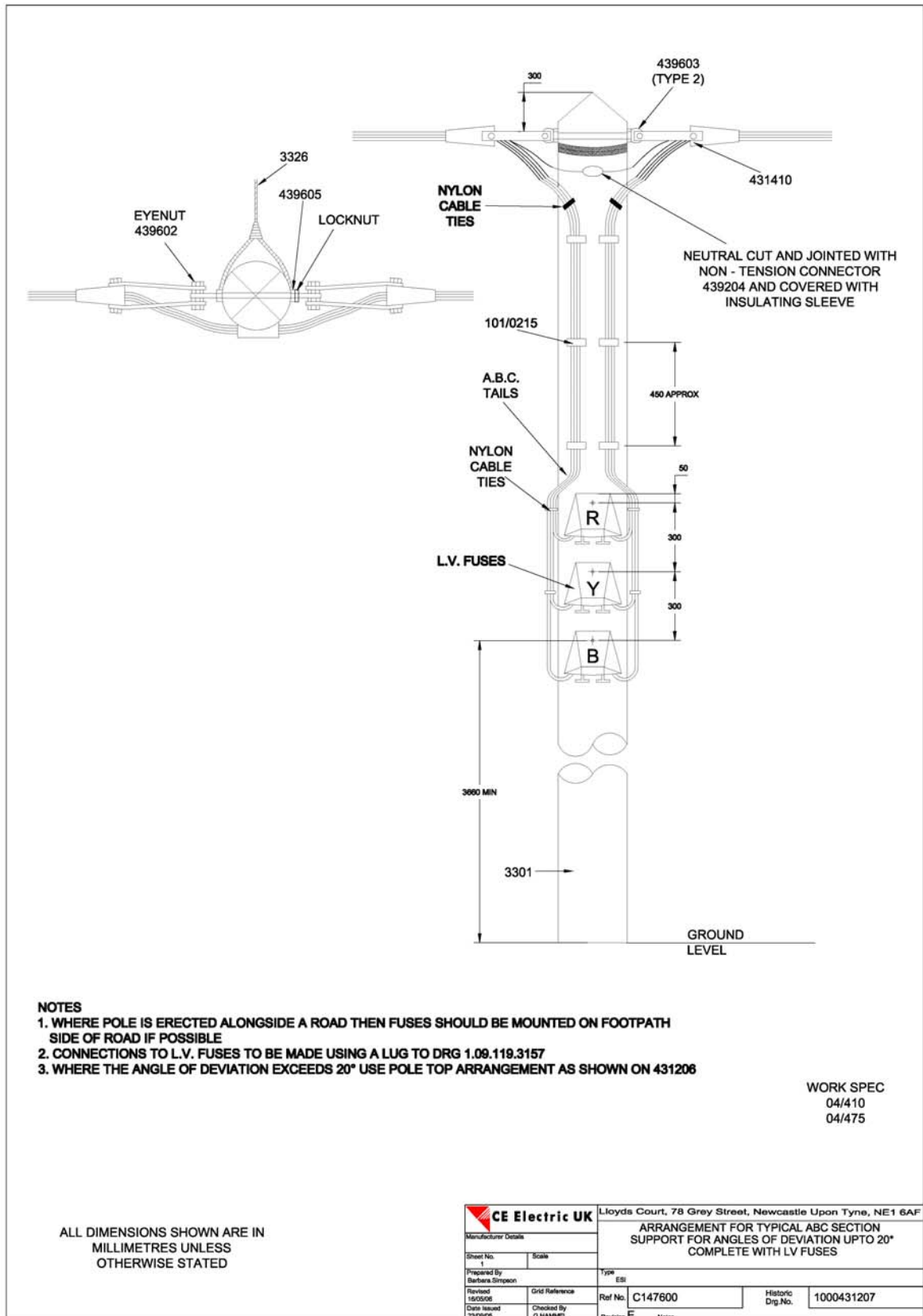
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Eye Bolts, M20, Type 2	439603	2	368059
Washer, Square/Curved	434605	4	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Anchor Clamp 2 x 35mm ²	431410	1	221085
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Section Angle support 20° to 90° Deviation

Drawing Number 1000431206

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MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Eye Bolts, M20, Type 2	439603	1	368059
Eye Nut, M20	439602	1	368025
Locknut M20	-	1	378915
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
<i>Requirements for LV Section Fuses (were req)</i>			
Non-Tension Joint Ucan 1.2	439204	1	268433
Insulated Sleeve	-	1	262210
Cable Cleats 4 x 35 mm ² ABC	101/0215	10	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	10	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	10	244797
Nylon Cable Ties	-	6	354111
Screw Nails, No 10 x 50 mm	-	20	371738
Lucy PC Fuse Units (Plastic)		3	184115
Cartridge Fuses (State Size)		3	To Suit
Tape, PVC, Self Adhesive	-	-	350483
<i>Tape, Identification, 12 mm</i>			
Red	-	-	350642
Yellow	-	-	352686
Blue	-	-	350835
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Section Angle support Up to 20° Deviation Complete with Section Fuses

Drawing Number 1000431207

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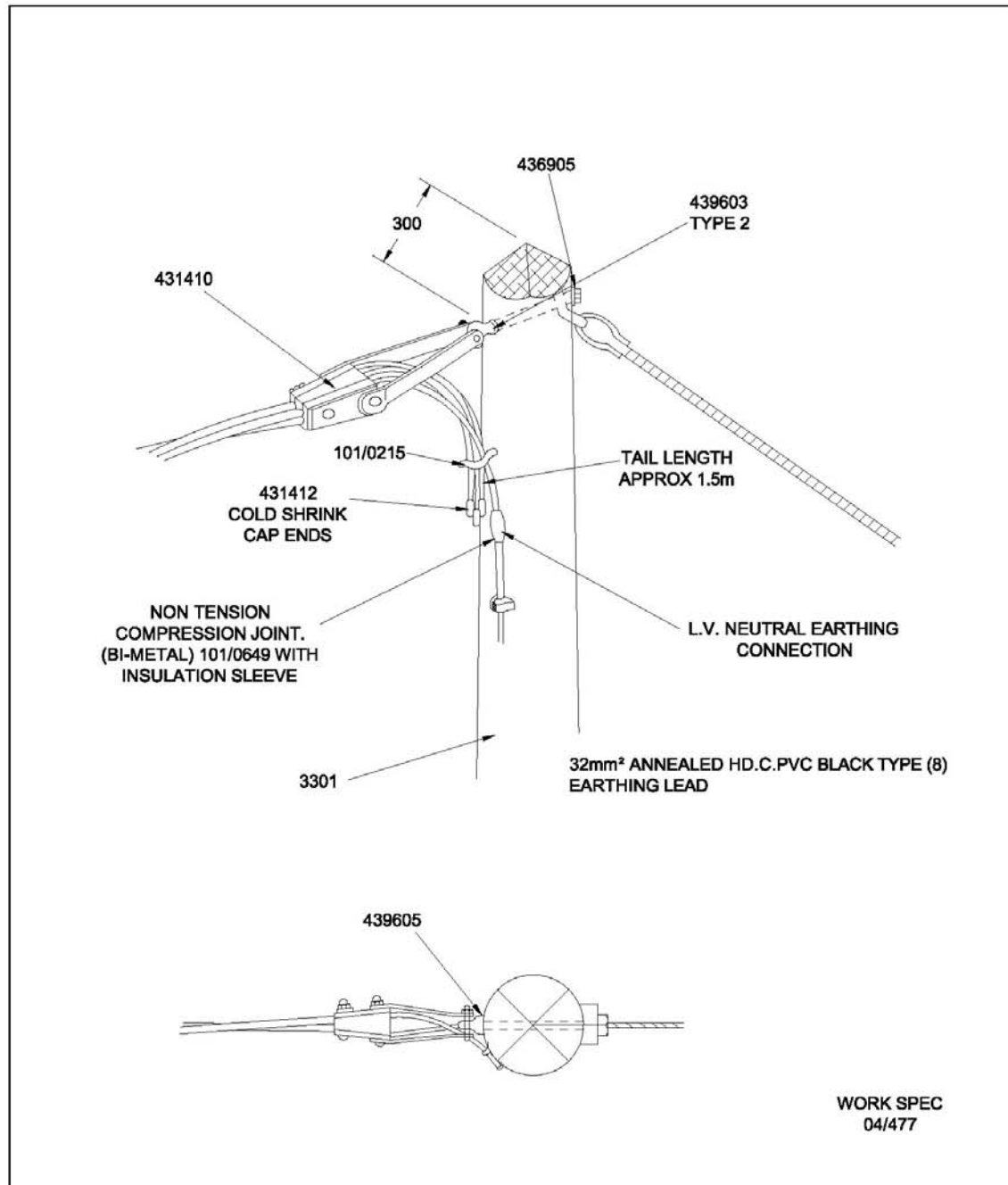
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
See drawing 1091193304 for details of Open Wire side of Pole			
Eye Bolts, M20, Type 2	439603	1	368059
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Insulation Piercing Connectors	431414 sht 3	4	242024
PVC Protective Sleeving	431411	1	265774
Cable Cleats 4 x 35 mm ² ABC	101/0215	3	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	3	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	3	244797
Nylon Cable Ties	-	6	354111
Screw Nails, No 10 x 50 mm	-	20	371738
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
For PME earthing if Required			
Insulation Piercing Connector	431414 sht 2	1	240090
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical Transition support between Open Wire and ABC

Drawing Number 1000431208

Document reference		NSP/004/041				
Version:-	4.0	Date of Issue:-	Sept 10	Page	51	of 77



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC TERMINAL SUPPORT			
Sheet No. 1	Scale	Type ESI			
Prepared By Barbara Simpson		Grid Reference			
Revised 16/05/06	Checked By G.HAMMEL	Ref No.	C319031	Historic Drg.No.	1000431209
Date Issued 23/09/05		Revision	E	Notes	

Document reference	NSP/004/041					
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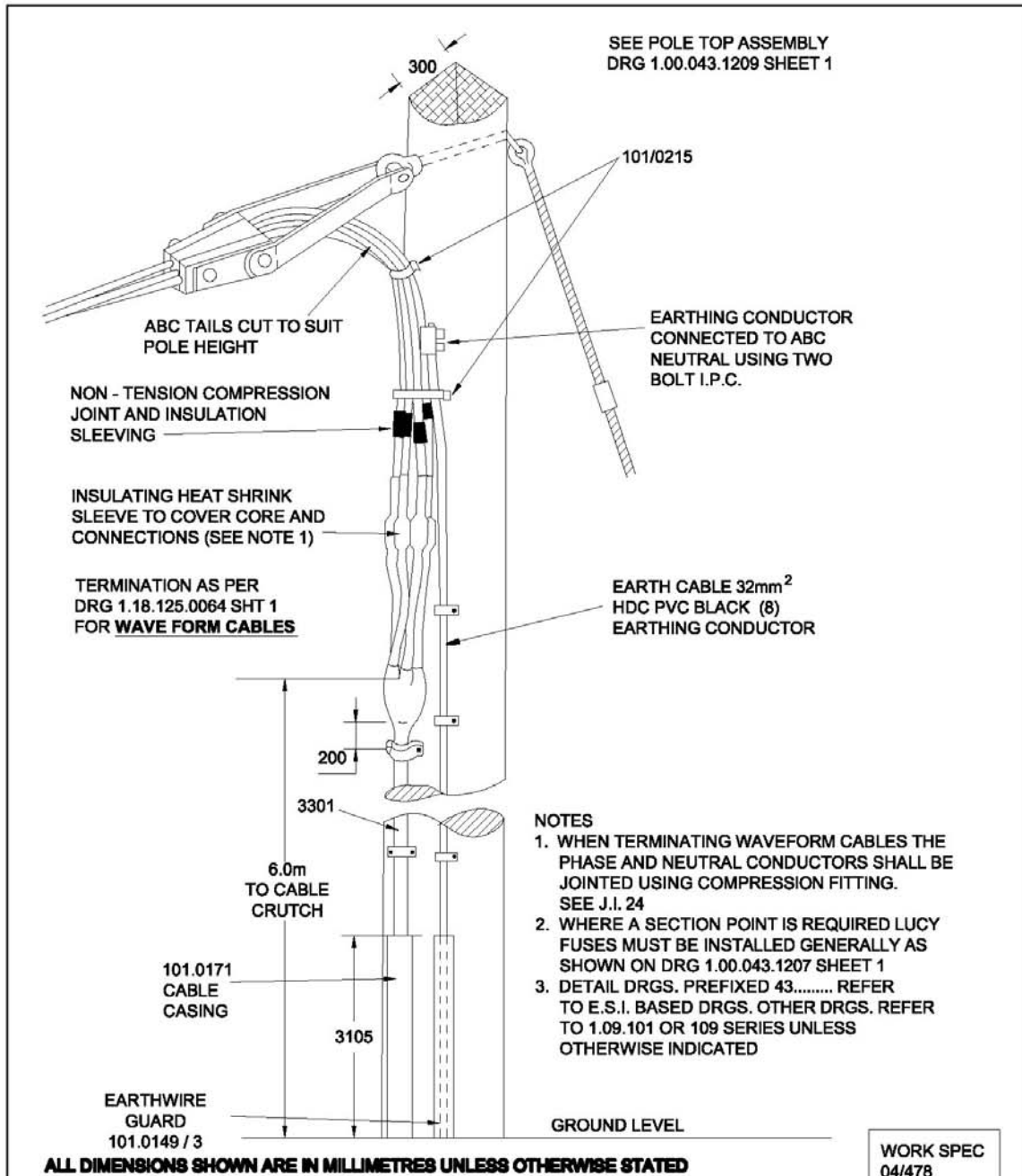
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
See drawing 1091193304 for details of Open Wire side of Pole			
Eye Bolts, M20, Type 2	439603	1	368059
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Cable Cleats 4 x 35 mm ² ABC	101/0215	2	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	2	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	2	244797
Nylon Cable Ties	-	6	354111
Screw Nails, No 10 x 50 mm	-	20	371738
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777
Or			
Screw Nails, Hardened	-	2	371738
For PME earthing if Required			
Non Tension Joint	101/0649	1	246932
Cable Joint Insulating Sleeve	-	1	262210
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Terminal Support

Drawing Number 1000431209

Document reference NSP/004/041		Date of Issue:- Sept 10		Page 53	of 77
Version:- 4.0					



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC TERMINAL SUPPORT WITH HEAT SHRINK CABLE TERMINATION			
Sheet No. 1	Scale	Type ESI			
Prepared By Barbara Simpson		Ref No. C147603			
Revised 18/05/08	Grid Reference	Historic Drq.No.		1000431210	
Date issued 23/09/05	Checked By G HAMMEL	Revision H		Notes	

Document reference		NSP/004/041				
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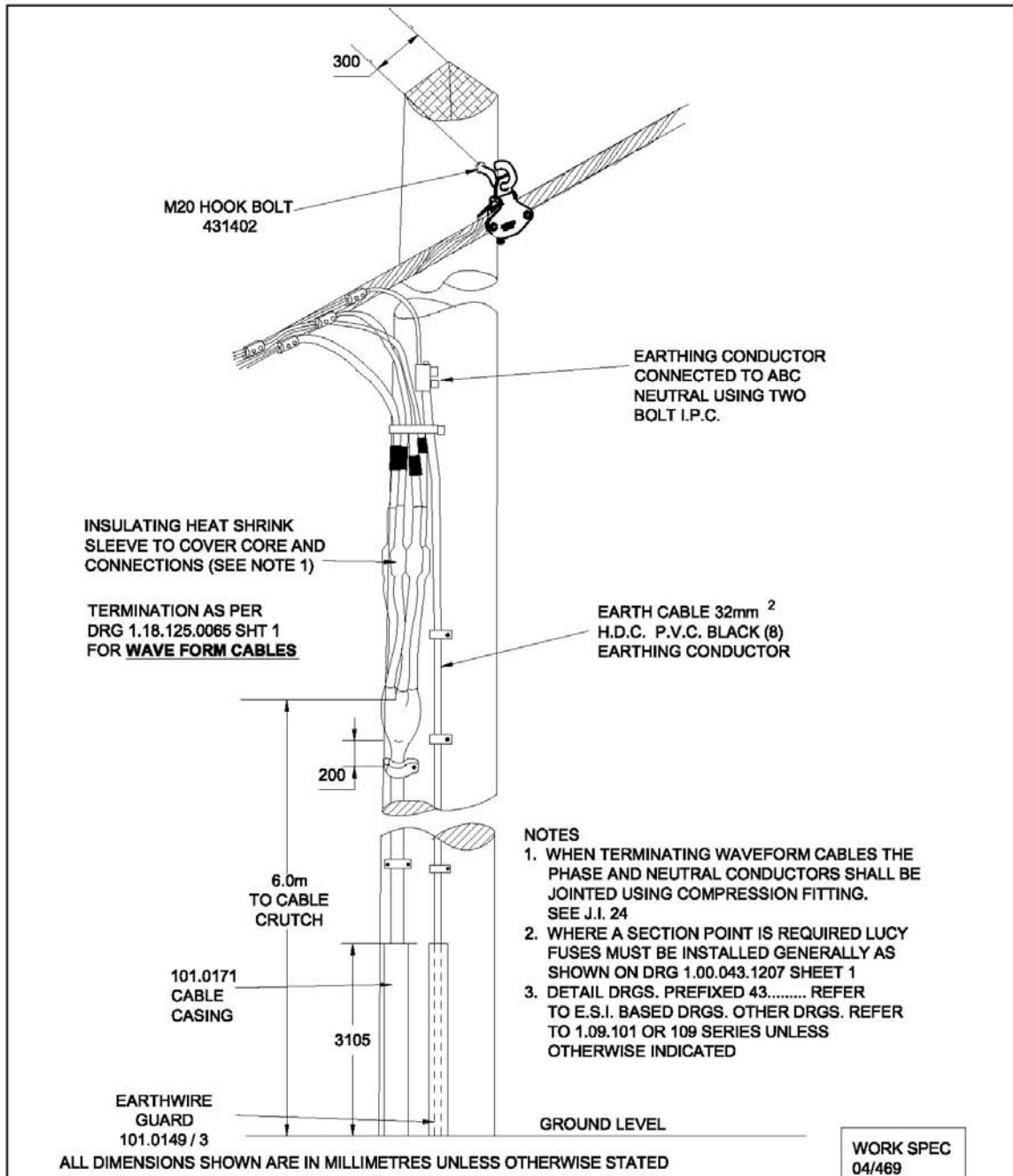
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Eye Bolts, M20, Type 2	439603	1	368059
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Cable Cleats 4 x 35 mm ² ABC	101/0215	5	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	5	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	5	244797
Cable Joint Insulating Sleeve	-	1	262210
Cable Guard - Fibre Glass	101/0171	1	To Suit
Screw Nails, No 10 x 50 mm	-	10	371738
Waveform Cable Termination Kit - 70 - 95mm	1.18.125.0064	1	086645
Waveform Cable Termination Kit - 120 - 300mm	1.18.125.0064	1	086660
Compression connectors to suit cable combination	-	-	To Suit
Insulation Piercing Connector	431414 sht 2	1	262070
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738
For PME earthing if Required			
Non Tension Joint	101/0649	1	246932
Cable Joint Insulating Sleeve	-	1	262210
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Terminal Support with Heat Shrink Cable Termination

Drawing Number 1000431210

Document reference		NSP/004/041				
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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC TEE - OFF FROM INTERMEDIATE SUPPORT WITH HEAT SHRINK CABLE TERMINATION			
Sheet No. 2	Scale				
Prepared By Barbara.Simpson		Type ESI			
Revised 18/05/06	Grid Reference	Ref No.	C147605	Historic Drg.No.	1000431210
Date Issued 23/09/05	Checked By G HAMMEL	Revision	D	Notes	

Document reference	NSP/004/041					
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MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Cable Cleats 4 x 35 mm ² ABC	101/0215	5	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	5	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	5	244797
Cable Joint Insulating Sleeve	-	1	262210
Cable Guard - Fibre Glass	101/0171	1	To Suit
Screw Nails, No 10 x 50 mm	-	10	371738
Waveform Cable Termination Kit - 70 - 95mm	1.18.125.0064	1	086645
Waveform Cable Termination Kit - 120 - 300mm	1.18.125.0064	1	086660
Compression connectors to suit cable combination	-	-	To Suit
Insulation Piercing Connector	431414 sht 2	1	262070
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738
For PME earthing if Required			
Non Tension Joint	101/0649	1	246932
Cable Joint Insulating Sleeve	-	1	262210
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Intermediate Support with Heat Shrink Cable Termination

Drawing Number 1000431210 sht2

Document reference		NSP/004/041				
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NOTES:

- WHERE A SECTION POINT IS REQUIRED, LUCY FUSES MUST BE INSTALLED GENERALLY AS SHOWN ON DRG No 1.00.043.1207
- DETAIL DRAWINGS PREFIXED 43..... REFER TO E.S.I. BASED DRAWINGS AND OTHER DETAIL DRAWINGS REFER TO 1.09.119 SERIES

WORK SPEC
04/479

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC TEE - OFF FROM INTERMEDIATE SUPPORT			
Sheet No. 1	Scale				
Prepared By Barbara Simpson		Type ESI			
Revised 18/05/06	Grid Reference	Ref No. C147606	Historic Drg.No.	1000431211	
Date Issued 01/09/05	Checked By G HAMMEL	Revision C Notes			

Document reference		NSP/004/041					
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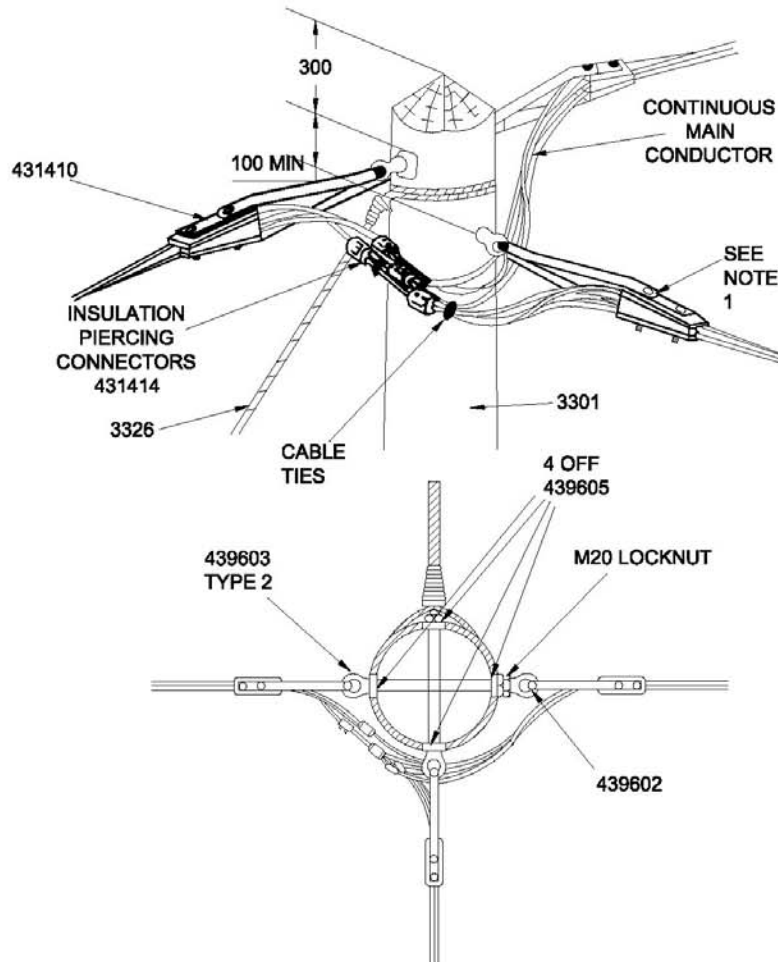
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Locknut M20	-	1	378915
Eyebolt M20	439602	1	368025
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Insulation Piercing Connector	431414 sht 2	4	241996
Nylon Cable Ties	-	4	354094
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Tee-Off From Intermediate Support

Drawing Number 1000431211 sht1

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NOTES

- 1, WHERE A SECTION POINT IS REQUIRED. LUCY FUSES MUST BE INSTALLED GENERALLY AS SHOWN ON DRAWING No 1.00.043.1207
2. DETAIL DRAWINGS PREFIXED 43..... REFER TO E.S.I. BASED DRAWINGS, AND OTHER DETAILED DRAWINGS REFER TO 1.09.119. SERIES

WORK SPEC
04/480

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL ABC TEE - OFF FROM SECTION SUPPORT			
Sheet No. 1	Scale				
Prepared By Barbara.Simpson		Type ESI			
Revised 18/05/06	Grid Reference	Ref No.	C319030	Historic Drg.No.	1000431212
Date Issued 03/03/05	Checked By G.HAMMEL	Revision	C	Notes	

Document reference		NSP/004/041					
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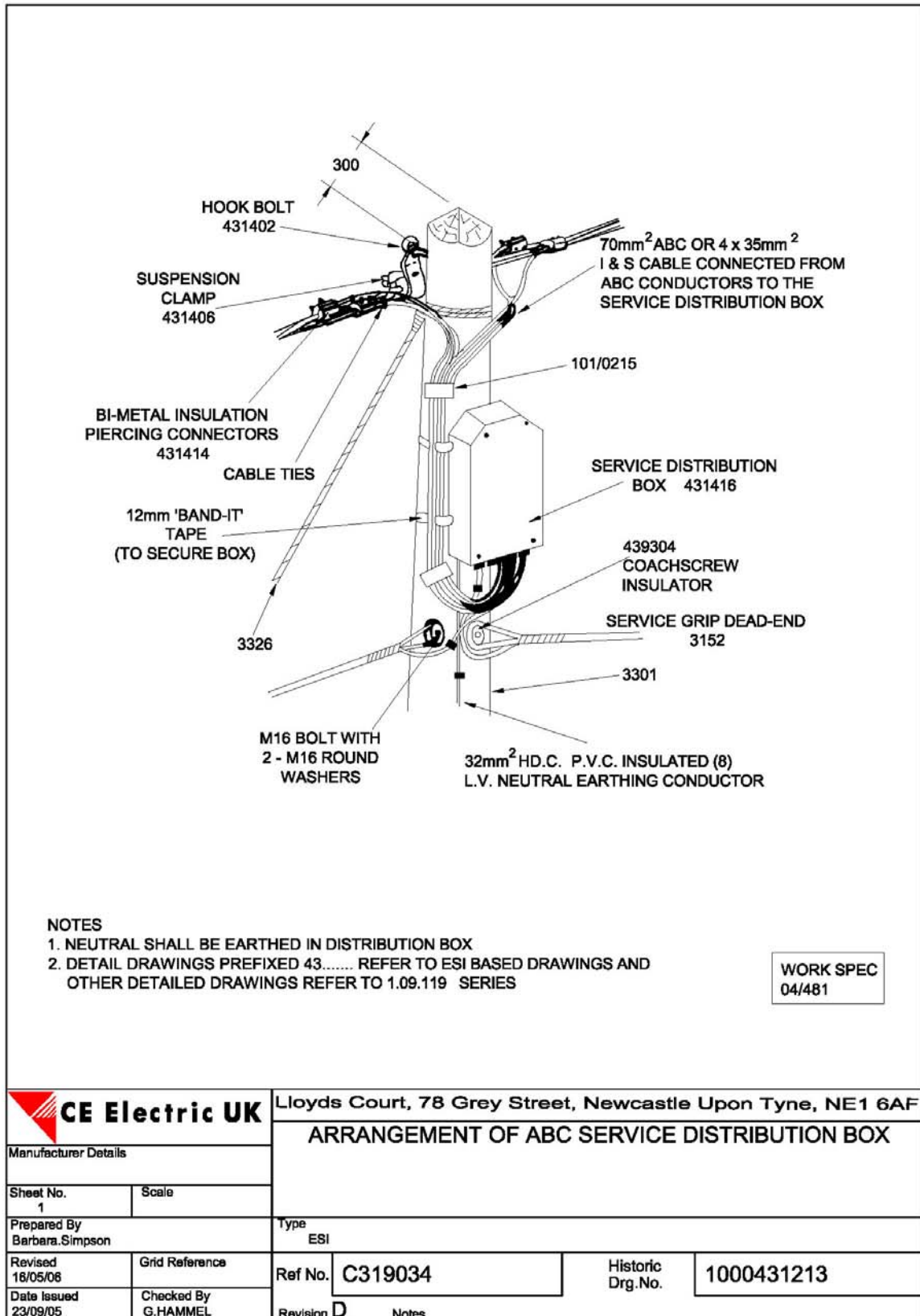
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Eye Bolts, M20, Type 2	439603	1	368059
Locknut M20	-	1	378915
Eyenuit M20	439602	1	368025
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Insulation Piercing Connector	431414 sht 2	4	241996
Nylon Cable Ties	-	4	354094
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738
<i>Stay Components</i>			
Bolt M20 x 220mm	-	1	378756
Washer, Square/Curved	434605	2	368078
Stay dead end	0541 item 1	3	255217
Pole Top dead end	0541 item 2	1	250209
Stay Thimble	9109	1	244922
Stay Plate	9110	1	237124
Stay Wire - 7/4.00mm Grade 1150		To suit	229609
Stay Insulator (type 1)	9107	1	248232
Stay Rod	9101		231477
Stay Block	9103 sht 2		235124

Arrangement of Typical ABC Tee-Off Section Support

Drawing Number 1000431212 sht1

Document reference		NSP/004/041				
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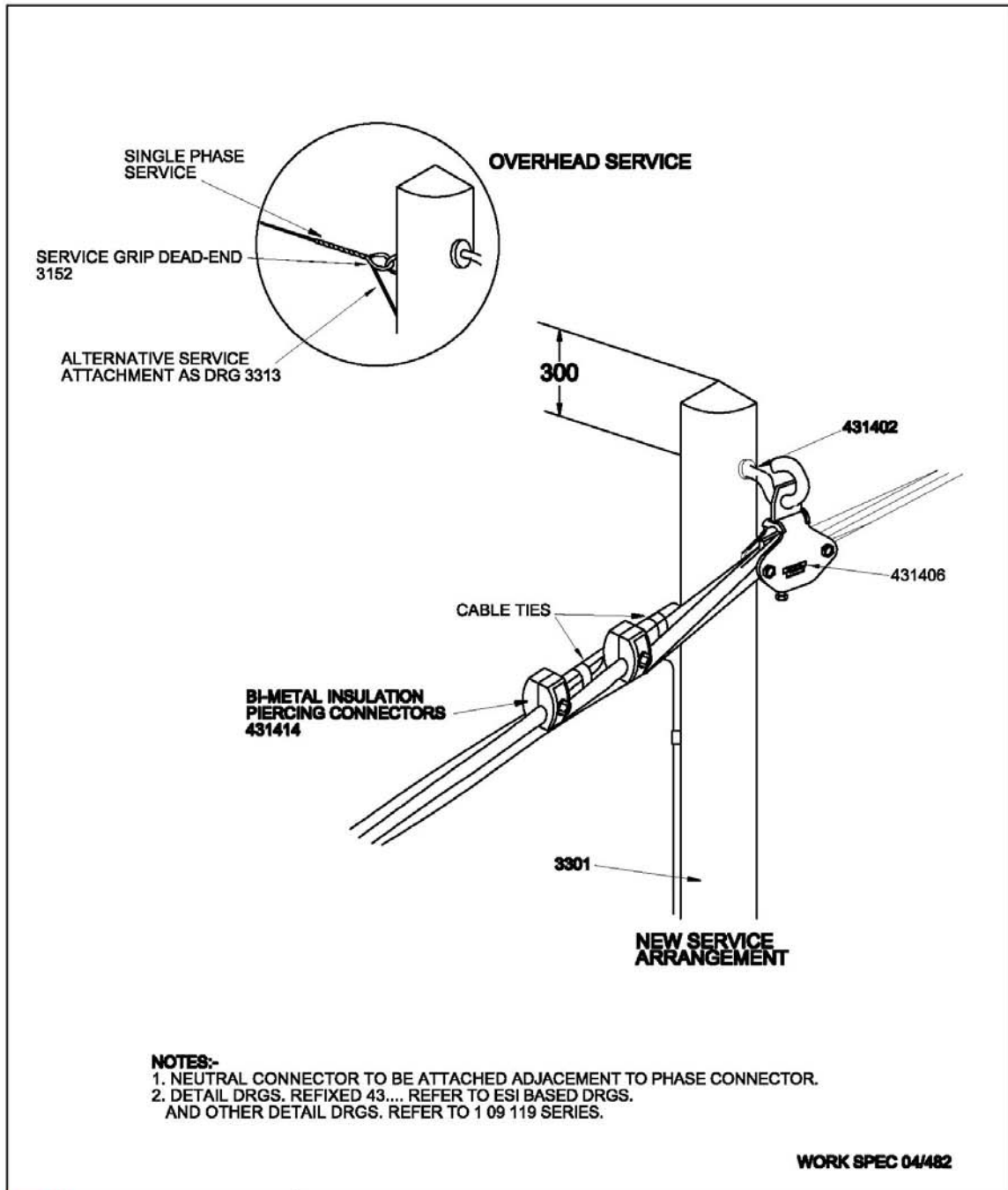
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Insulation Piercing Connector	431414 sht 2	4	241996
Nylon Cable Ties	-	10	354094
Service Distribution Box	431416	1	262028
4 x 35mm ABC		2m	022855
4 x 35mm I & S Single Core Cables		8m	028563
12 mm 'Band It' Tape			365120
Cable Cleats 4 x 35 mm ² ABC	101/0215	3	244763 or
Cable Cleats 4 x 70 mm ² ABC	101/0215	3	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	3	244797
Single Way, Cable Cleats	3112	As Req	248815
Nails, Screw No 10 x 2 in		As Req	371738
Coachscrew Service Insulator	9305	As Req	253868
Single Cable Cleat	3112	As Req	248815
Service Grip Dead Ends	3152	As Req	To Suit
Or			
Anchor Clamp 4 x 35mm ABC - Reduced Tension 1.4kN	-	2	221069
Anchor Clamp 2 x 35mm ABC - Reduced Tension 1.4kN	-	As Req	221077
Eye Bolts, M20, Type 2	439603	As Req	368059
Washer, Square/Curved	439605	As Req	368075
For PME earthing if Required			
Insulation Piercing Connector	431414 sht 2	1	240090
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of Typical Service Distribution Box

Drawing Number 1000431213 sht1

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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT FOR TYPICAL SINGLE PHASE SERVICE FROM ABC SUPPORT			
Sheet No. 1	Scale N.T.S.				
Prepared By Janis.Webb		Type ESI			
Revised 10/01/07	Grid Reference	Ref No.	C147611	Historic Drg.No.	1000431214
Date Issued 23/09/05	Checked By G HAMMEL	Revision	E	Notes	

Document reference		NSP/004/041					
Version:-	4.0	Date of Issue:-	Sept 10	Page	64	of	77

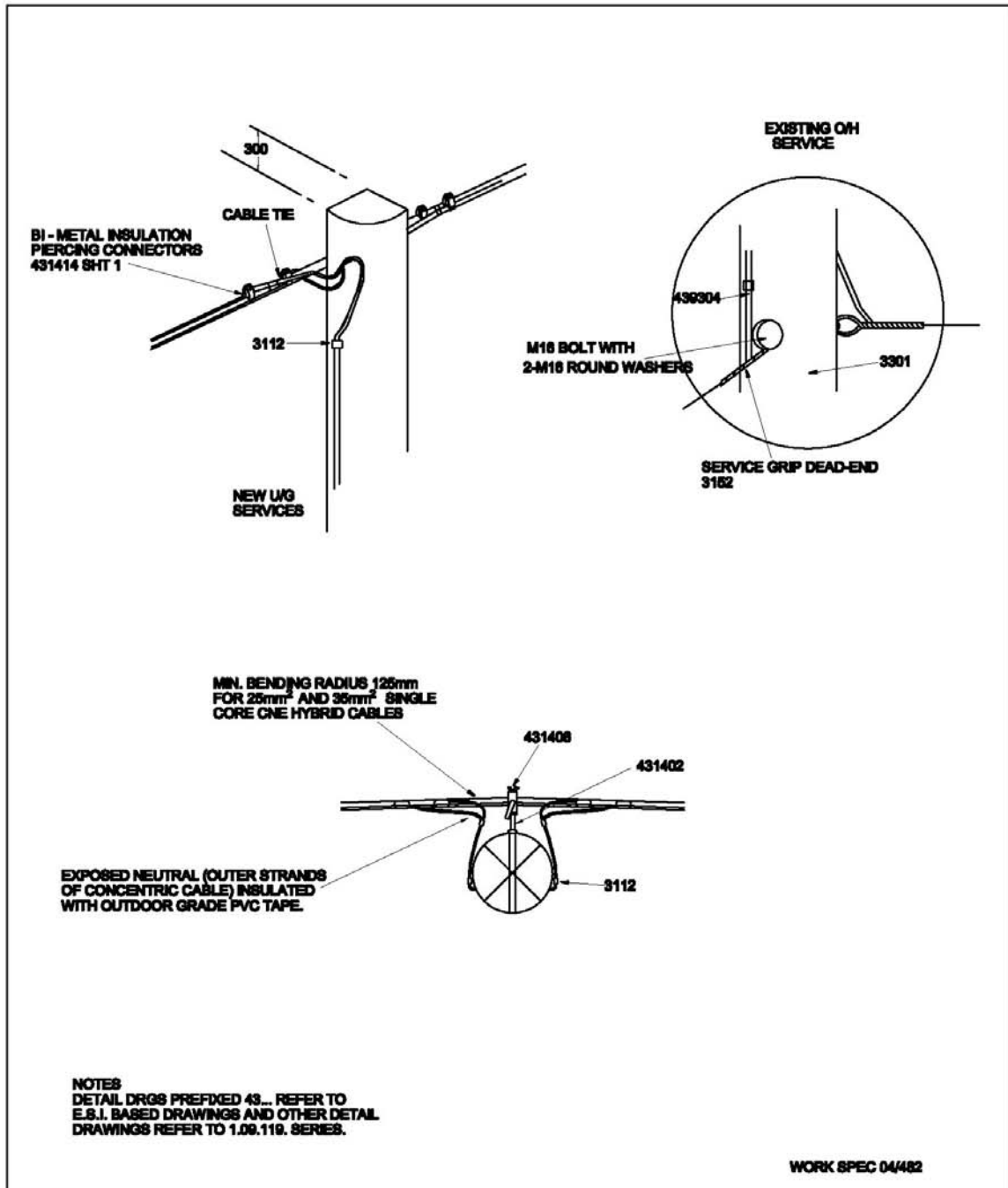
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Insulation Piercing Connector	431414 sht 1	2	242132
Nylon Cable Ties	-	2	354094
Coachscrew Service Insulator	9305	1	253868
Single Cable Cleat	3112	1	248815
Service Grip Dead Ends	3152	As Req	To Suit
Or			
Anchor Clamp 2 x 35mm ABC - Reduced Tension 1.4kN	-	1	221077
Anchor Clamp 4 x 35mm ABC - Reduced Tension 1.4kN	-	2	221069
Eye Bolts, M20, Type 2	439603	1	368059
Washer, Square/Curved	439605	1	368075
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of Typical Single Service Phase Service from ABC Intermediate Support

Drawing Number 1000431214 sht1

Document reference		NSP/004/041				
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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
		ARRANGEMENT FOR TYPICAL MULTIPLE SERVICE TEE OFF FROM A.B.C.			
Manufacturer Details					
Sheet No. 1	Scale				
Prepared By Barbara Simpson		Type ESI			
Revised 18/05/08	Grid Reference	Ref No.	C317362	Historic Drg.No.	1000431215
Date issued 23/09/05	Checked By G.HAMMEL	Revision	C	Notes	

Document reference		NSP/004/041					
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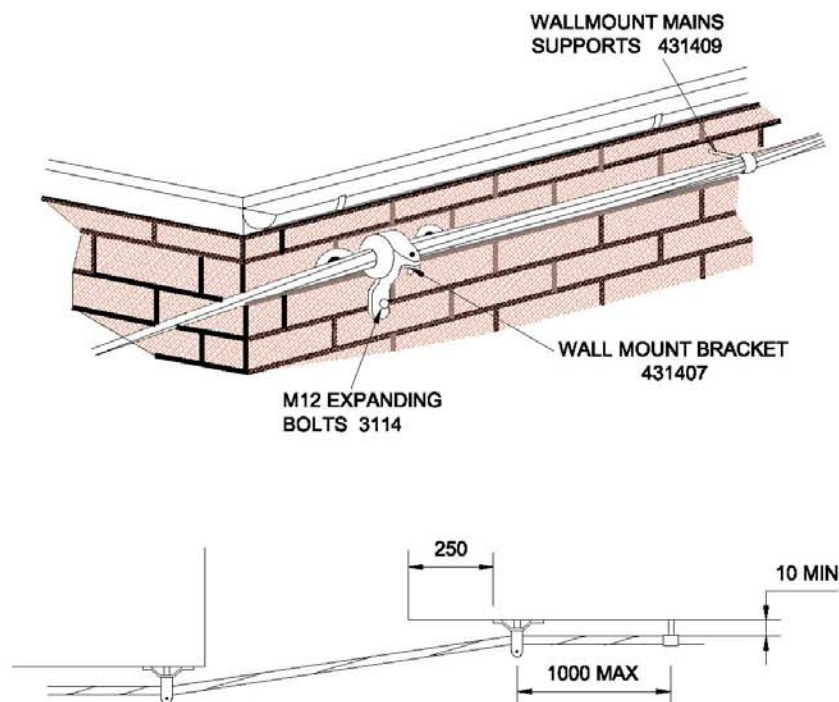
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Insulation Piercing Connector	431414 sht 1	As req	242132
Nylon Cable Ties	-	As req	354094
Coachscrew Service Insulator	9305	As req	253868
Single Cable Cleat	3112	As req	248815
Service Grip Dead Ends	3152	As Req	To Suit
Or			
Anchor Clamp 2 x 35mm ABC - Reduced Tension 1.4kN	-	2	221077
Anchor Clamp 4 x 35mm ABC - Reduced Tension 1.4kN	-	2	221069
Eye Bolts, M20, Type 2	439603	2	368059
Washer, Square/Curved	439605	2	368075
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of Typical Multiple Service Take off from ABC Intermediate Support

Drawing Number 1000431215 sht1

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NOTES:
DETAIL DRAWINGS PREFIXED 43 REFER TO ESI BASED DRAWINGS AND OTHER
DETAILED DRAWINGS REFER TO 1.08.119 SERIES

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		TYPICAL ATTACHMENT FOR ABC ON AND BETWEEN BUILDINGS			
Sheet No. 1	Scale				
Prepared By Barbara.Simpson		Type ESI			
Revised 17/05/06	Grid Reference	Ref No.	C316865	Historic Drg.No.	1000431217
Date Issued 01/08/05	Checked By G.HAMMEL	Revision	B	Notes	

Document reference		NSP/004/041					
Version:-	4.0	Date of Issue:-	Sept 10	Page	68	of	77

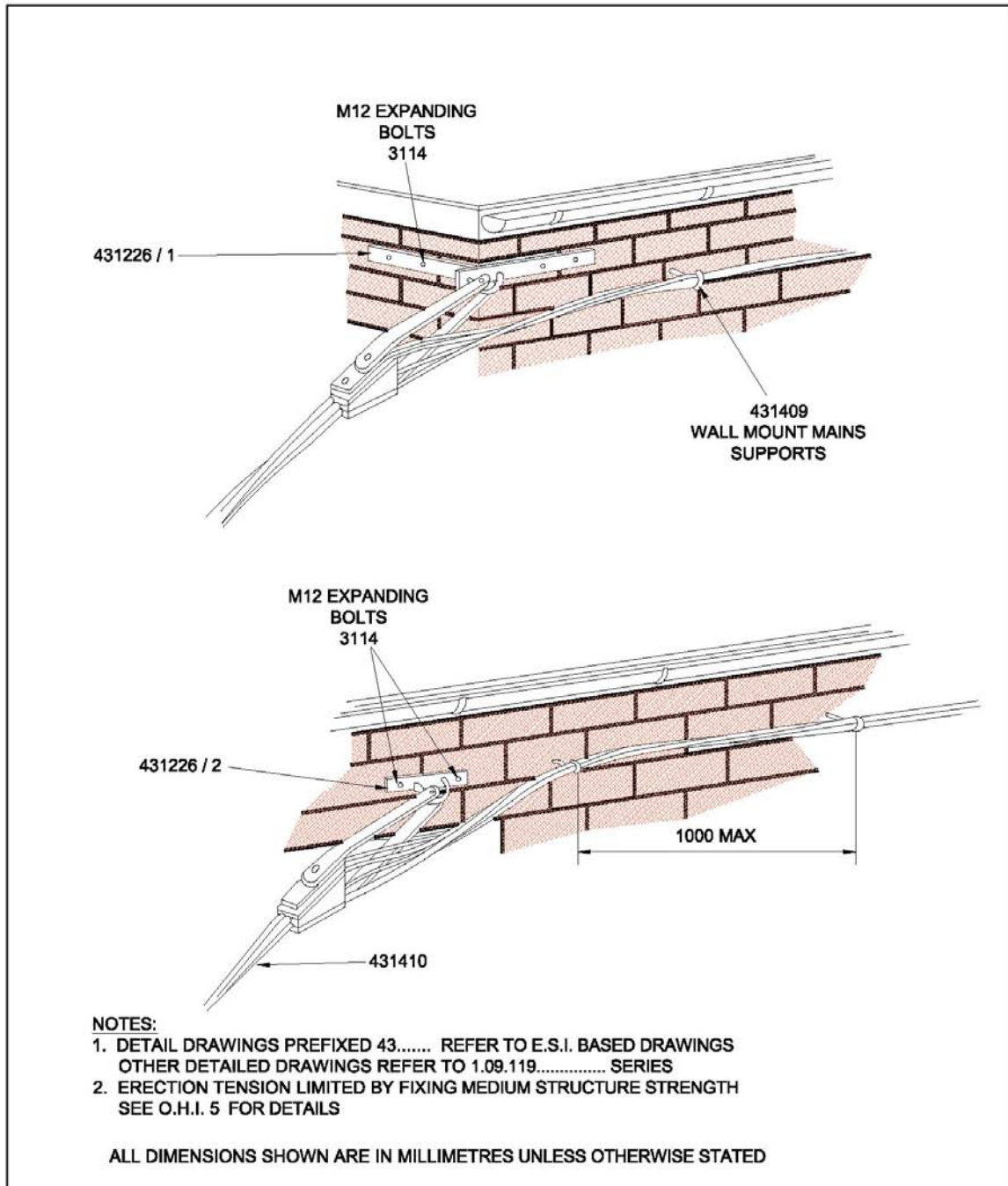
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Foundation Block	439103 sht2	1	235124
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of Typical Multiple Service Take off from ABC Intermediate Support

Drawing Number 1000431216 sht1

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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT OF TYPICAL ATTACHMENTS TO BUILDING			
Sheet No. 2	Scale				
Prepared By Barbara Simpson		Type ESI			
Revised 18/05/06	Grid Reference	Ref No.	C316867	Historic Drg.No.	1000431217
Date Issued 01/08/05	Checked By G.HAMMEL	Revision	D	Notes	

Document reference		NSP/004/041					
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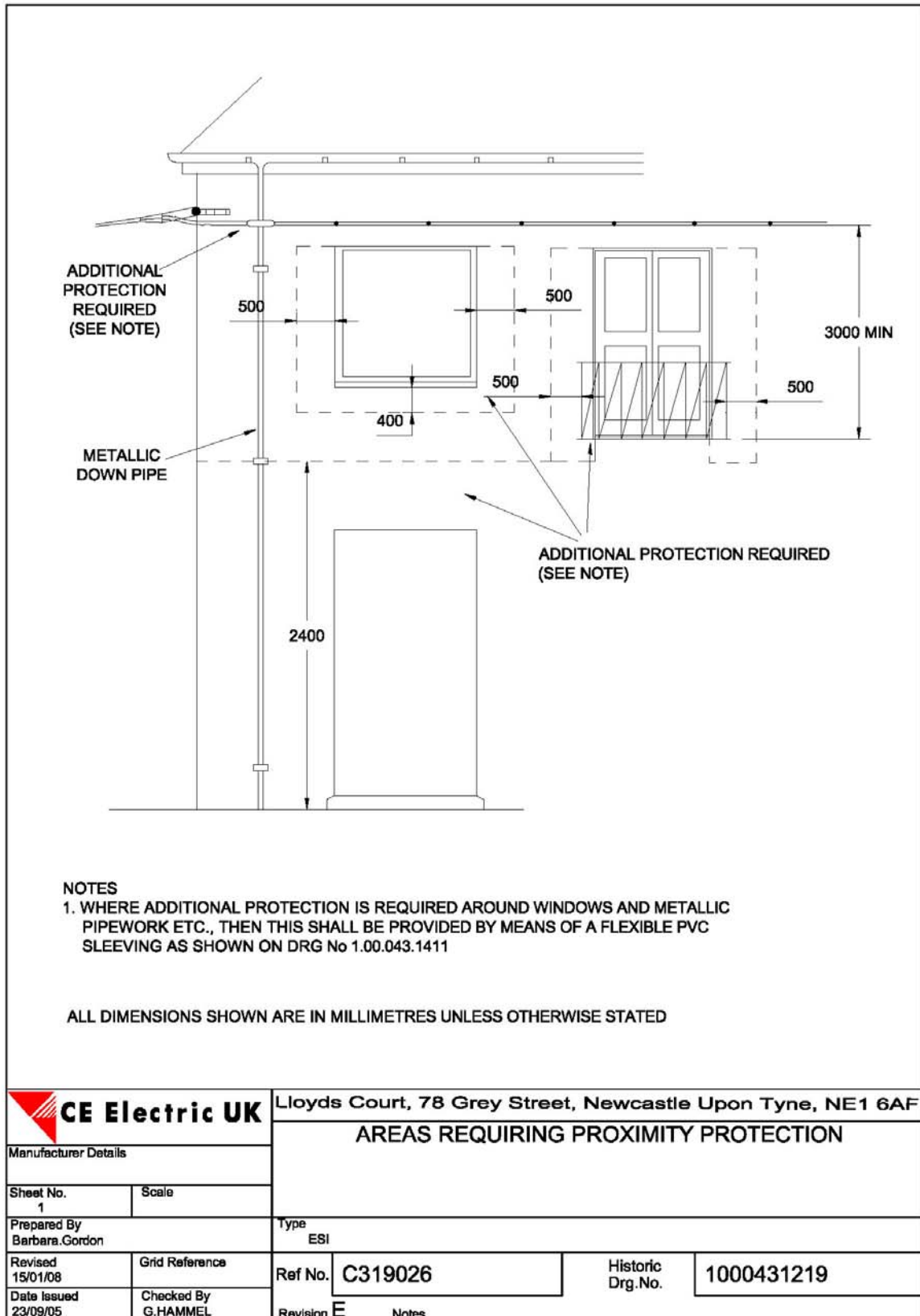
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
House service bracket	439510	1	241341or
House service bracket	439514	1	242236
Support Wall mains 100mm stand off	431409	1	262051
Wall Mount bracket	431407	1	To Suit
Fixings used with above brackets			
Bolts, M12 Expanding	-	2	371121

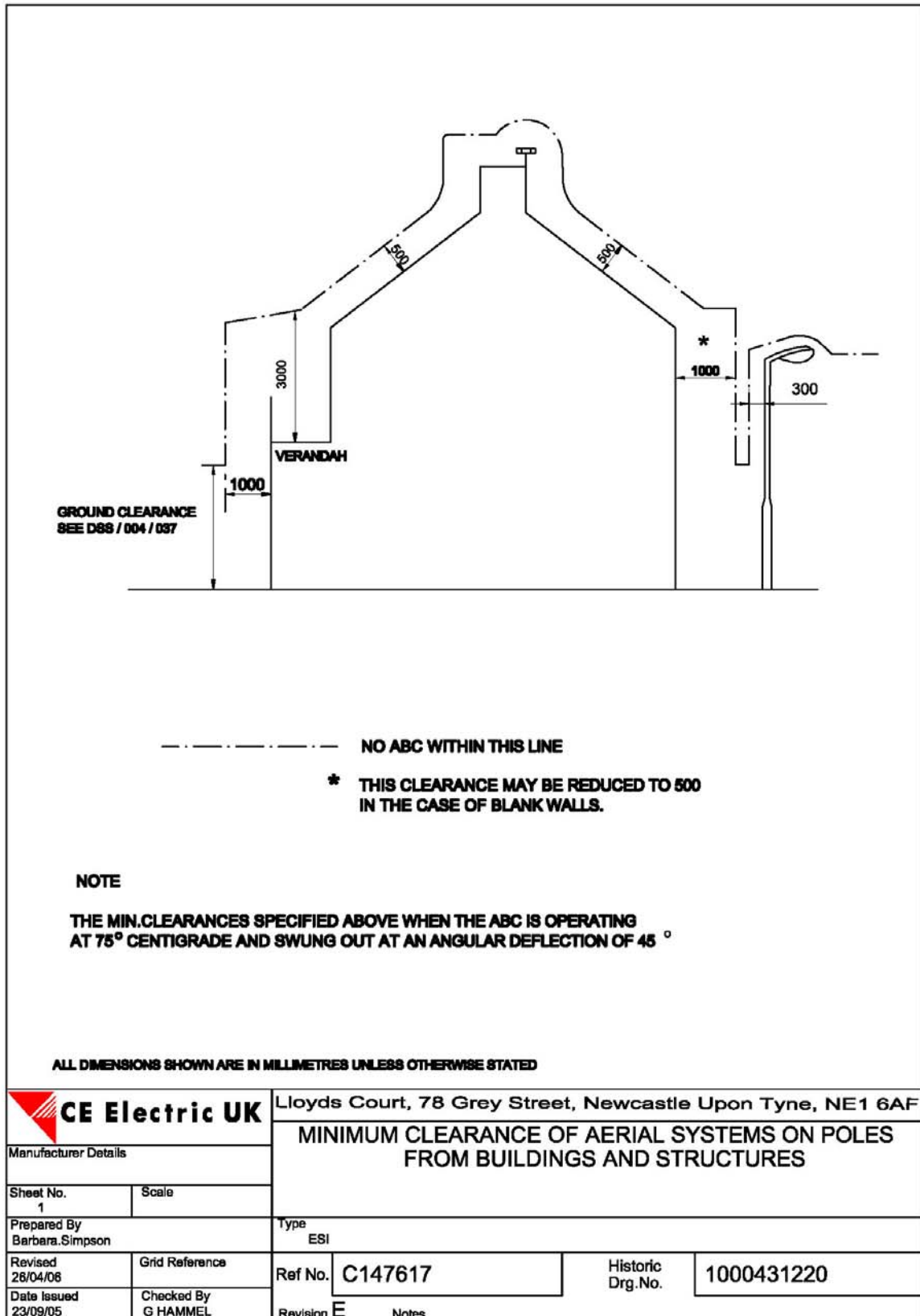
Typical Attachment for ABC on and between buildings

Drawing Number 1000431217 sht1 & 2

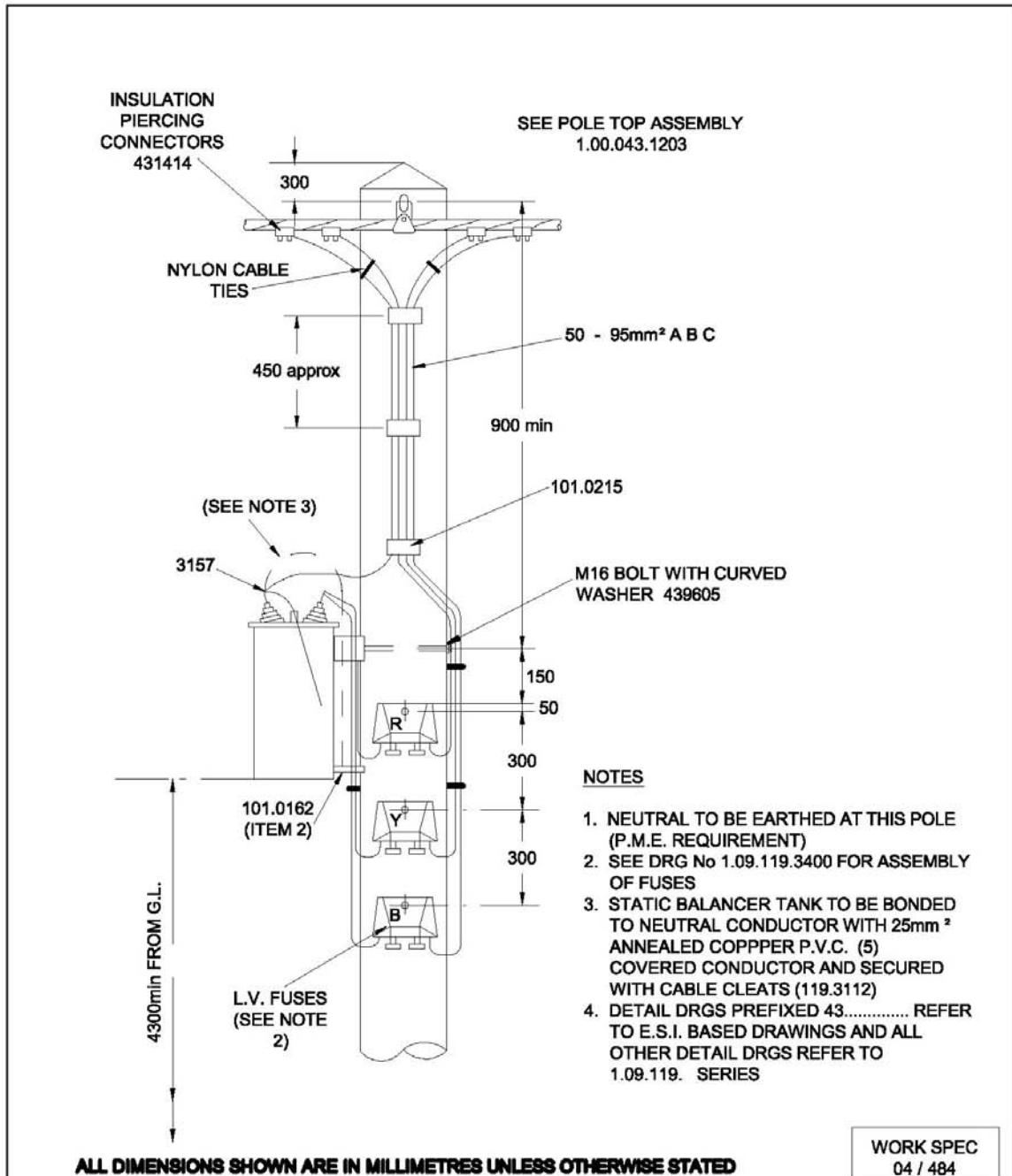
Document reference		NSP/004/041					
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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT OF STATIC BALANCER ON WOOD POLE			
Sheet No. 1	Scale				
Prepared By Barbara.Simpson		Type ESI			
Revised 28/04/06	Grid Reference	Ref No.	C316866	Historic Drg.No.	1000431222
Date issued 01/08/05	Checked By G.HAMMEL	Revision	C	Notes	

Document reference		NSP/004/041					
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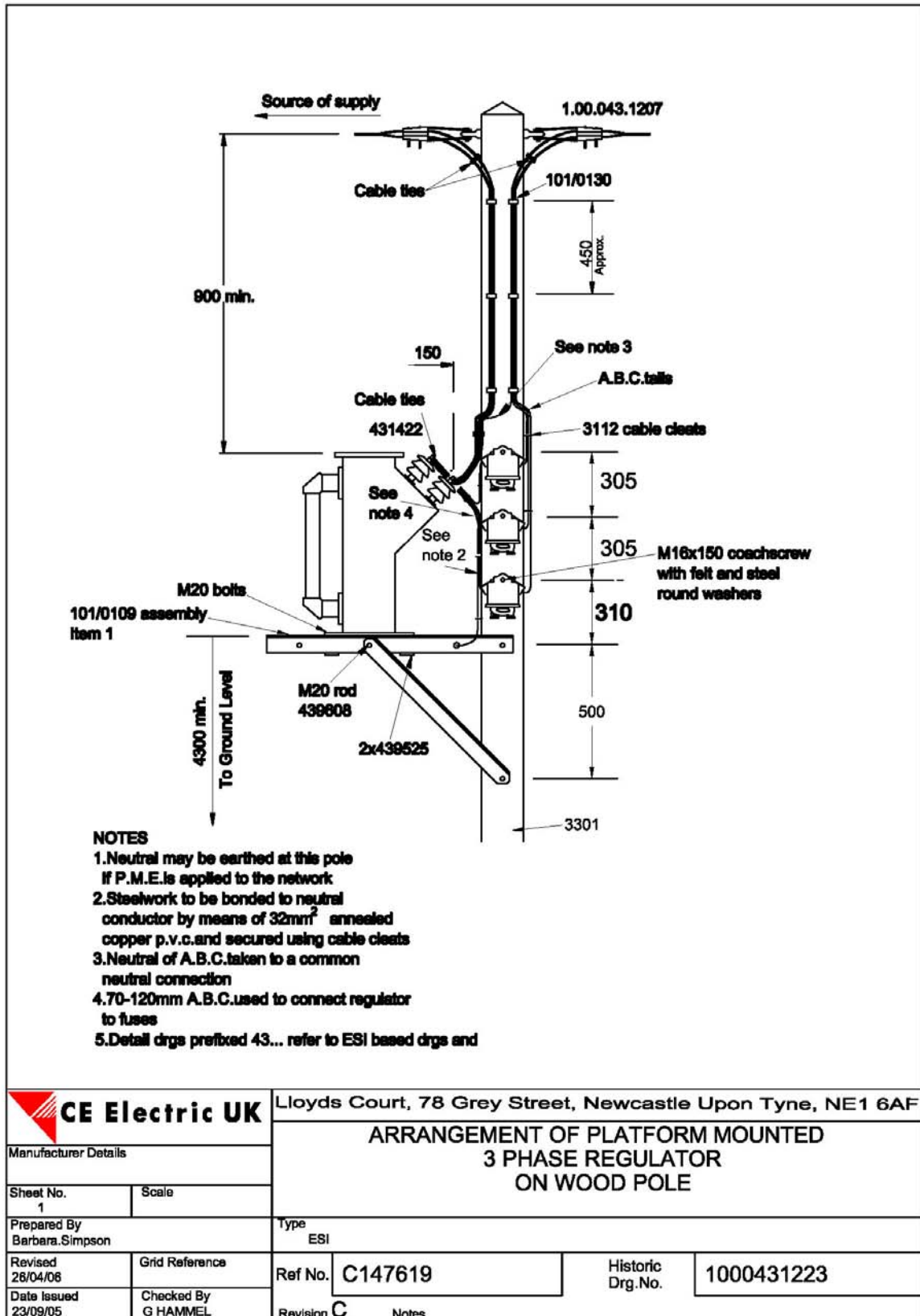
MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Hook Bolt, M20, 220mm	431402	1	265596
Washer, Square/Curved	434605	2	368078
Suspension Roller Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431406	1	242217
Bolt Galv M20 x 300	-	1	378794
Washer Galv, Round	-	1	375616
Washer, Square/Curved	434605	1	368078
Bolt, earthing	101/0162	1	370491
Insulation Piercing Connector	431414 sht 2	4	241996
Nylon Cable Ties	-	8	354094
Lugs, Compression BL5	101/0237	4	To Suit
Lugs Compression, VPCL 1.1	101/0237	2	268838
4 x 70mm ABC		6m	226031
4 x 120mm ABC		6m	226021
Cable Cleats 4 x 70 mm ² ABC	101/0215	6	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	6	244797
Single Way, Cable Cleats	3112	8	248815
Nails, Screw No 10 x 2 in		20	371738
For PME earthing if Required			
Insulation Piercing Connector	431414 sht 2	1	240090
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of ABC Supplying a Static Balancer

Drawing Number 1000431222 sht1

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MATERIALS LIST

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Eye Bolts, M20, Type 2	439603	1	368059
Eye Nut, M20	439602	1	368025
Locknut M20	-	1	378915
Washer, Square/Curved	434605	2	368078
Anchor Clamp (Range taking) 4 x 35mm ² / 4 x 120mm ² & 2 x 35mm ²	431410	1	242359
Lucy PC Fuse Units (Plastic)		3	184115
Cartridge Fuses (State Size)		3	To Suit
Support Platform Kit	101/0612 sht 8	2	245306
Bolt Galv M20 x 300	-	1	378794
Rod, Tie M20, 330	439608 item 4	1	375828
Bolts M20 x 140mm	-	4	378101
Washer Galv, Round	-	8	375616
Washer, Square/Curved	434605	1	368078
Bolt, earthing	101/0162	1	370491
Lugs, Compression BL5	101/0237	8	To Suit
Lugs Compression, VPCL 1.1	101/0237	2	268838
4 x 70mm ABC		6m	226031
4 x 120mm ABC		6m	226021
Cable Cleats 4 x 70 mm ² ABC	101/0215	6	244778 or
Cable Cleats 4 x 120mm ² ABC	101/0215	6	244797
Single Way, Cable Cleats	3112	8	248815
Nails, Screw No 10 x 2 in		20	371738
For PME earthing			
Insulation Piercing Connector	431414 sht 2	1	240090
32 mm ² HDC Black PVC Insulated (6)		10m	224755
Single Core Cable Cleats		10	248815
Nails, Screw, No 10 x 2 in		10	371738
Earth Lead Casing		1	235548
Earth Electrode		1	354249
Earth Clamp		1	354338
<i>Signs & Notices</i>			
Number Plate (select cat numbers to suit)	0228 sht3	To Suit	To Suit
Base Plate	0228 sht3	1	243258
Safety Sign	0229 sht1	1	363318
Fibre Washer 9/16 in dia x 1/4 thick	-	6	374581
Fibre Washer 9/16 in dia x 1/16 thick	-	6	374577
Screws 1 in x No 8 brass	-	6	375777 Or
Screw Nails, Hardened	-	2	371738

Arrangement of Platform Mounted 3 Phase Regulator on Wood pole

Drawing Number 100043123 sht1

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